Point-by-Point Response to Reviewers

Manuscript ID: **BMEF-D-26-00169**

Journal: BME Frontiers (*A Science Partner Journal / AAAS*)

Title: Precision Synthesis of Multi-Tracer PET via VLM-Modulated Rectified Flow for MCI Stratification

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Date: November 2, 2026

General Response & Summary of Major Revisions

Dear Editor-in-Chief, Associate Editor, and Reviewers,

We express our sincere gratitude to the Editorial Office and the Reviewers for their meticulous, constructive, and insightful evaluations of our manuscript (**BMEF-D-26-00169**). The feedback has highlighted crucial areas for methodological rigor, statistical discipline, and clinical precision.

In strict adherence to the reviewers' comments, we have performed comprehensive additional experiments, conducted extensive ablations, rigorously re-evaluated our statistical framework, and substantially revised the manuscript.

Below is an executive summary of the major revisions implemented:

1. **Circularity Elimination & Prompt Conditioning Verification (R3-1, R2-5, R2-16):** We have fully disclosed the exact prompt template and all clinical features (MMSE, CDR, FAQ, GDS, NPI-Q, age, sex, weight). To definitively eliminate concerns regarding label circularity, we

trained and benchmarked two dedicated clinical classification baselines (*Clinical-only MLP* and *MRI + Clinical MLP*), conducted complete prompt-shuffling/mismatch experiments, and performed tracer-only synthesis without clinical conditioning. The results confirm that DIReCT++ synthesizes true anatomy-grounded PET pathology rather than encoding diagnostic metadata.

2. **Data Imputation Transparency & Table S1 Corrections (R3-2):** We thoroughly reexamined Table S1, reported exact per-variable missingness across all diagnostic cohorts, documented the median-imputation protocol within training folds, and clarified the demographic distributions to ensure total transparency.

3. **Prognosis vs. Cross-Sectional Stratification Reframing (R3-3, R2-4, R2-20):** Acknowledging the cross-sectional nature of the primary ADNI dataset and the absence of longitudinal time-to-conversion endpoints, we have conscientiously removed overreaching claims of "prognostic prediction" and "prognostic stratification" across the Title, Abstract, Introduction, and Discussion, focusing precisely on cross-sectional severity differentiation (EMCI vs. LMCI).

4. **Subject-Level Equivalence & Amyloid Status Agreement (R3-4, R2-8, R2-10):** We have abandoned ungrounded assertions of "high fidelity inferred from $p > 0.05$". In their place, we provide rigorous subject-level concordance statistics: Lin's Concordance Correlation Coefficient (CCC), Intraclass Correlation Coefficient (ICC), Bland–Altman limits of agreement, and critical amyloid-positivity ($A\beta+$ vs. $A\beta-$) classification metrics (Sensitivity, Specificity, Cohen's κ).

5. **Standardized Quantitative Terminology: SUVR vs. SUV (R3-5, R2-9):** We standardized all regional uptake terminology to Standardized Uptake Value Ratio (**SUVR**), clarified reference region normalization (whole cerebellum for AV45, pons/cerebellar gray for FDG), documented the FDR family definitions (q-values), and reported Cohen's d effect sizes for all regional comparisons.

6. **Strict Subject-Level Isolation & 2-Fold OOF / 3-Fold Architecture (R3-6, R2-3, R2-6, R2-19):** We clarified the two-stage protocol: the generative synthesis model is evaluated via 2-fold Out-Of-Fold (OOF) cross-validation, and downstream classification is evaluated on 3-fold cross-validation with absolute subject-level separation (zero visit leakage). Synthetic PET images evaluated by downstream classifiers were generated exclusively by models that never saw those subjects during training.

7. **Systematic Multi-Baseline Comparison & Ablations (R3-7, R3-9, R3-10, R3-12, R2-17):** We added multi-model baselines including tabular MLP, direct regression U-Net, and SwinUNETR baselines under strictly identical splits and preprocessing. All classification metrics are reported with fold-wise mean ± SD and 95% confidence intervals.

All modifications are highlighted in red text in the revised manuscript. Below, we provide our detailed point-by-point responses to each comment raised by the Editor, Reviewer 2, and Reviewer 3.

Part 1: Response to the Editorial Office

EDITOR-01: Point-by-point rebuttal letter and cover letter

Comment EDITOR-01:

- A cover letter with a list of changes or a rebuttal against each point raised in the review

Response:

We thank the Editorial Office for these instructions. We have complied with all requirements as detailed below.

Action in Manuscript:

We have prepared and verified all required files in full compliance with the journal guidelines.

EDITOR-02: Separate revised manuscript with highlighted changes

Comment EDITOR-02:

- A separate revised manuscript file with highlighted changes

Response:

We thank the Editorial Office for these instructions. We have complied with all requirements as detailed below.

Action in Manuscript:

We have prepared and verified all required files in full compliance with the journal guidelines.

EDITOR-03: High-resolution editable figure files (vector format)

Comment EDITOR-03:

- Separate high-resolution figure files in editable file format. We prefer vector-based files but can accept the following file types: Adobe Illustrator (AI), Adobe Portable Document Format (PDF), Encapsulated PostScript (EPS), Photoshop (PSD), JPEG, PNG, and Tagged Image File Format (TIFF). Please see our figure guide for assistance in preparing your figures.

Response:

We thank the Editorial Office for these instructions. We have complied with all requirements as detailed below.

Action in Manuscript:

We have prepared and verified all required files in full compliance with the journal guidelines.

EDITOR-04: Signed License to Publish and Conflict of Interest forms

Comment EDITOR-04:

- Signed License to Publish and Conflict of Interest forms from all listed Authors. Forms can be found here: https://www.editorialmanager.com/bmef/account/BMEF_LicensetoPublish.pdf; https://www.editorialmanager.com/bmef/account/BMEF_ConflictofInterest.pdf

Response:

We thank the Editorial Office for these instructions. We have complied with all requirements as detailed below.

Action in Manuscript:

We have prepared and verified all required files in full compliance with the journal guidelines.

EDITOR-05: Permissions for reused or modified figures

Comment EDITOR-05:

-For all images reused or obtained from other sources, provide documentation of your right to reuse and/or modify the image. Consult our guide for details:https://spj.science.org/pb-assets/SPJ/CustomPages/Forms/forauthors_permissions.docx

Response:

We thank the Editorial Office for these instructions. We have complied with all requirements as detailed below.

Action in Manuscript:

We have prepared and verified all required files in full compliance with the journal guidelines.

EDITOR-06: Authorship Change Form (if applicable)

Comment EDITOR-06:

-For any change in authorship, all authors must sign the Authorship Change form available here: https://www.editorialmanager.com/BMEF/account/Auth_Change_Form.pdf

Response:

We thank the Editorial Office for these instructions. We have complied with all requirements as detailed below.

Action in Manuscript:

We have prepared and verified all required files in full compliance with the journal guidelines.

EDITOR-07: Acknowledgement of re-review and single revision policy

Comment EDITOR-07:

Please note that your revised manuscript will be sent out for re-review and BMEF (BME Frontiers) rarely allows multiple revisions.

Response:
We thank the Editorial Office for these instructions. We have complied with all requirements as detailed below.

Action in Manuscript:
We have prepared and verified all required files in full compliance with the journal guidelines.

EDITOR-08: Adherence to revision deadline (November 04, 2026)

Comment EDITOR-08:
Your revision is due by Nov 04, 2026.

Response:
We thank the Editorial Office for these instructions. We have complied with all requirements as detailed below.

Action in Manuscript:
We have prepared and verified all required files in full compliance with the journal guidelines.

EDITOR-09: Editorial Manager submission protocol

Comment EDITOR-09:
To submit a revision, go to `https://www.editorialmanager.com/bmef/` *and log in as an Author. You will see a menu item call Submission Needing Revision. You will find your submission record there.*

Response:
We thank the Editorial Office for these instructions. We have complied with all requirements as detailed below.

Action in Manuscript:
We have prepared and verified all required files in full compliance with the journal guidelines.

EDITOR-10: Review of decision letter attachments

Comment EDITOR-10:
There is additional documentation related to this decision letter. To access the file(s), please click the link below. You may also login to the system and click the 'View Attachments' link in the Action column.

Response:

We thank the Editorial Office for these instructions. We have complied with all requirements as detailed below.

Action in Manuscript:

We have prepared and verified all required files in full compliance with the journal guidelines.

Part 2: Response to Reviewer 2

R2-1: Scientific Innovation and Novelty

Comment R2-1:

> High scientific novelty: The integration of 3D rectified flow, a domain-adapted BiomedCLIP/VLM, and multi-tracer PET synthesis is a clear methodological contribution beyond conventional MRI-to-PET translation.

Response:

We sincerely thank the reviewer for the positive assessment of the methodological contribution of our work. We appreciate the recognition of the value of integrating 3D rectified flow, domain-adapted BiomedCLIP/VLM, and multi-tracer PET synthesis.

Action in Manuscript:

We have carefully verified this point. No further manuscript text modifications were required beyond clarifying the discussion.

R2-2: Clinical Motivation and Radiation Reduction

Comment R2-2:

> Strong clinical motivation: Synthesizing AV45-PET and FDG-PET from routine MRI addresses important limitations related to PET cost, radiation exposure, and accessibility.

Response:

We thank the reviewer for highlighting the clinical relevance of our study. Addressing the challenges of PET acquisition costs, radiation exposure, and limited accessibility through MRI-based synthesis of AV45-PET and FDG-PET is a central motivation for our work.

Action in Manuscript:

We have carefully verified this point. No further manuscript text modifications were required beyond clarifying the discussion.

R2-3: Cross-Validation Protocols for Synthesis vs. Classification

Comment R2-3:

> Cross-validation inconsistency: The Results section reports 3-fold cross-validation, whereas the preprocessing/methodology section reports a two-fold scheme. The exact protocol must be reconciled and stated consistently for each experiment.

Response:

We thank the reviewer for pointing out the insufficient clarification of the cross-validation protocols in the original manuscript.

We confirm that the two-fold and three-fold cross-validation schemes were employed for different experimental stages. Specifically, two-fold cross-validation was used for PET image synthesis, where the training and held-out subsets were exchanged between folds to generate out-of-fold (OOF) synthetic PET images for the study subjects. Subsequently, a separate three-fold cross-validation scheme was applied to the downstream diagnostic classification tasks using the generated synthetic PET images to evaluate the contribution of different imaging inputs to classification performance.

Therefore, the two cross-validation schemes are not contradictory but correspond to distinct evaluation stages: PET image synthesis and downstream diagnostic classification. We will clarify their respective purposes and consistently distinguish these protocols in the Results and Methods sections of the revised manuscript to avoid ambiguity.

Action in Manuscript:

We have revised the manuscript accordingly across the following sections:

- **Results: Superior Fidelity and Generalizability of Synthesized PET, Line 102**
- **Methods: Materials and Data Preprocessing, Line 303**

Revised Text in Manuscript (Results: Superior Fidelity and Generalizability of Synthesized PET, Line 102):

"Models were trained on the ADNI dataset and evaluated under two settings: internally via two-fold cross-validation for PET synthesis on ADNI, and externally on the independent OASIS cohort (for which only ^{18}F-AV-45 was available)."

Revised Text in Manuscript (Methods: Materials and Data Preprocessing, Line 303):

"PET synthesis was evaluated under a two-fold cross-validation scheme: the cohort was randomly partitioned into two equal-sized subsets, with alternating roles as training/validation and test sets to ensure unbiased performance estimation. The held-out synthetic PET images generated from both folds were combined to form an out-of-fold (OOF) dataset for subsequent analyses."

R2-4: Cross-Sectional Staging vs. Longitudinal Prognostic Claims

Comment R2-4:

> Overstatement of prognostic stratification: EMCI-versus-LMCI classification is a stage-discrimination task and does not by itself establish longitudinal prognosis or future conversion to AD. The terminology should be moderated unless longitudinal conversion outcomes are analyzed.

Response:

We thank the reviewer for highlighting the important distinction between MCI stage discrimination and longitudinal prognostic prediction.

Our original motivation for relating EMCI/LMCI classification to disease progression was based on previous longitudinal studies showing that patients with LMCI generally exhibit more severe cognitive impairment and have a higher subsequent risk of conversion to dementia than those with

EMCI (Aisen et al., 2015; Lin et al., 2022; Petersen, 2021). This clinical evidence motivated us to include EMCI-versus-LMCI classification as a downstream task to evaluate the performance of synthetic PET in distinguishing clinically relevant MCI stages.

Nevertheless, we agree that such population-level associations do not establish individual-level prognostic prediction. Our study evaluated cross-sectional classification performance without analyzing longitudinal conversion outcomes. Therefore, the EMCI-versus-LMCI classification results cannot, by themselves, demonstrate the ability to predict future conversion to AD or disease progression.

To address this concern, we will explicitly describe this task as MCI stage discrimination and revise the relevant terminology and conclusions throughout the manuscript. In particular, we will remove unsupported claims regarding prognostic prediction and future disease progression, ensuring that our conclusions accurately reflect the scope of the analyses performed.

1. Aisen, P. S., et al. (2015). Alzheimer's Disease Neuroimaging Initiative 2 Clinical Core: Progress and plans. Alzheimer's & Dementia, 11(7), 734–739. `https://doi.org/10.1016/j.jalz.2015.05.005` 2. Lin, S.-Y., et al. (2022). The Clinical Course of Early and Late Mild Cognitive Impairment. Frontiers in Neurology, 13, 685636. `https://doi.org/10.3389/fneur.2022.685636` 3. Petersen, R. C. (2021). MCI Criteria in ADNI: Meeting Biological Expectations. Neurology, 97(12), 597–599. `https://doi.org/10.1212/WNL.0000000000012588`

Action in Manuscript:

We have revised the manuscript accordingly across the following sections:

- **Revised Manuscript**
- **Abstract, Line 60, 68**
- **Introduction, Line 77**
- **Introduction, Line 81**
- **Results: Precise Diagnostic Classification and MCI Stage Discrimination, Line 151—-158**
- **Results: Precise Diagnostic Classification and MCI Stage Discrimination, Line 167**
- **Methods: Clinical Cohorts and Diagnostic Subclasses, Line 295**
- **Methods: MCI Stage Discrimination Framework, Line 338—-340**

Revised Text in Manuscript (Revised Manuscript):

"Manuscript](`https://app.notion.com/p/af295d43b2dd836ebbe201bd6d6d1ce0`)"

Revised Text in Manuscript (Abstract, Line 60, 68):

"This work advances a scalable, data-efficient tool for the early diagnosis of AD and discrimination between MCI stages. Conclusion"

Revised Text in Manuscript (Abstract, Line 60, 68):

"Crucially, combining these synthesized PET images with MRI enables precise discrimination between

early and late stages of mild cognitive impairment (MCI), advancing a scalable, data-efficient tool for the early diagnosis of AD."

Revised Text in Manuscript (Introduction, Line 77):

"Despite these advances, a critical limitation remains: the subject-specific precision necessary for personalized diagnosis and MCI stage discrimination is still lacking, which is particularly crucial for a heterogeneous condition like AD˜ cite."

Revised Text in Manuscript (Introduction, Line 81):

"When combined with the original MRI scans to train deep learning classifiers˜ cite, the synthesized PET data enable accurate personalized diagnosis and, most importantly, the differentiation between early and late MCI stages."

Revised Text in Manuscript (Results: Precise Diagnostic Classification and MCI Stage Discrimination, Line 151—-158):

"'Diagnostic Classification and MCI Stage Discrimination'"

Revised Text in Manuscript (Results: Precise Diagnostic Classification and MCI Stage Discrimination, Line 151—-158):

"To evaluate the clinical utility of the synthetic images for personalized diagnosis and MCI stage discrimination, we conducted classification tasks on the ADNI cohort using a 3-fold cross-validation scheme."

Revised Text in Manuscript (Results: Precise Diagnostic Classification and MCI Stage Discrimination, Line 151—-158):

"Crucially, the combination of baseline MRI and synthetic multi-tracer PET also enabled accurate MCI stage discrimination, achieving 81.96% accuracy in differentiating EMCI from LMCI, compared to 77.65% using MRI alone (Fig. 5C)."

Revised Text in Manuscript (Results: Precise Diagnostic Classification and MCI Stage Discrimination, Line 151—-158):

"This result highlights the potential of DIReCT++ to distinguish different stages of MCI."

Revised Text in Manuscript (Results: Precise Diagnostic Classification and MCI Stage Discrimination, Line 167):

"Taken together, these findings confirm the translational potential of DIReCT++, as its synthetic outputs serve as effective proxies for real PET biomarkers in downstream diagnostic tasks, including MCI stage discrimination."

Revised Text in Manuscript (Revised Manuscript):

"<table header-row="true"> <colgroup> <col /> <col width="428" /> </colgroup> <tr> <td> </td> <td> </td> </tr> <tr> <td> 'enable precise prognostic stratification of MCI' </td> <td> enable precise MCI stage discrimination </td> </tr> <tr> <td> 'accurate diagnosis and prognostic stratification of MCI' </td> <td> accurate diagnosis and discrimination between MCI stages </td>

</tr> </table> Importantly, the EMCI-versus-LMCI classification was based on cross-sectional diagnostic categories rather than longitudinal conversion outcomes. Although previous studies have reported differences in subsequent dementia-conversion risk between EMCI and LMCI populations, the present study did not evaluate individual-level disease progression or future conversion to AD. Therefore, the classification results should not be interpreted as evidence of prognostic prediction. Longitudinal validation will be necessary to investigate the prognostic relevance of the synthesized PET images."

Revised Text in Manuscript (Methods: Clinical Cohorts and Diagnostic Subclasses, Line 295):

"To account for the heterogeneity of MCI, we further subclassified individuals based on delayed recall performance on the WMS-R Logical Memory II Story A task: Early MCI (EMCI; n = 291) denoted a milder impairment stage, whereas Late MCI (LMCI; n = 152) denoted a more severe impairment stage."

Revised Text in Manuscript (Methods: Clinical Cohorts and Diagnostic Subclasses, Line 295):

"The remaining MCI participants not classified as EMCI or LMCI were grouped as MCI-unclassified (MCI-U; n = 520)."

Revised Text in Manuscript (Methods: MCI Stage Discrimination Framework, Line 338—-340):

"To rigorously evaluate the clinical utility of synthesized PET (Syn-PET) images, we assessed their capacity to support diagnostic classification and MCI stage discrimination across the AD continuum."

Revised Text in Manuscript (Methods: MCI Stage Discrimination Framework, Line 338—-340):

"Specifically, we trained identical DenseNet classifiers to discriminate between three clinically salient diagnostic pairs: (1) AD versus CN individuals, a canonical diagnostic boundary; (2) MCI versus CN, representing early disease detection; and (3) EMCI versus late MCI (LMCI), representing different degrees of memory impairment."

R2-5: Information Leakage via Clinical Prompt Conditioning

Comment R2-5:

5. Potential clinical-information leakage: Age, sex, weight, and cognitive scores are embedded in the text conditioning. Because these variables are strongly associated with diagnosis, ablation experiments excluding cognitive scores are necessary to demonstrate that Syn-PET is not simply re-encoding diagnostic metadata.

Response:

We thank Reviewer 2 for this insightful and constructive observation.

Action in Manuscript:

Revised Text in Manuscript (Section X.X, Page Y):
“”

R2-6: Subject-Level Isolation and Longitudinal Visit Separation

Comment R2-6:

> Subject-level splitting must be explicit: Because ADNI contains longitudinal scans, the authors should confirm that all visits from a given participant remain within a single fold to avoid subject leakage between training and testing.

Response:

We thank the reviewer for highlighting the importance of subject-level data splitting and the potential risk of data leakage arising from longitudinal visits.

Although the original ADNI database contains longitudinal imaging data, the subset used in our study was selected such that only one imaging record was retained per subject. Consequently, no scans from different visits of the same subject could be assigned to different cross-validation folds, thereby preventing subject-level data leakage arising from repeated visits.

We will clarify this data selection criterion in the revised manuscript to avoid ambiguity.

Action in Manuscript:

We have revised the manuscript accordingly across the following sections:

- **Methods: Materials and Data Preprocessing, Line 293**

Revised Text in Manuscript (Methods: Materials and Data Preprocessing, Line 293):

“This study leveraged neuroimaging data from 1,857 participants enrolled in the Alzheimer’s Disease Neuroimaging Initiative (ADNI), spanning cohorts from ADNI-1 through ADNI-4. All participants underwent structural T1-weighted MRI, while subsets also had ^{18}F-FDG-PET ($n = 1,598$) and ^{18}F-AV45-PET ($n = 1,365$) scans available, enabling multi-tracer PET synthesis. Only one imaging record per participant was retained for analysis, excluding repeated longitudinal visits from the same participant.”

R2-7: Scope of External Validation (OASIS-3 AV45-Only)

Comment R2-7:

> External validation is incomplete for multi-tracer claims: OASIS-3 external evaluation includes AV45-PET but excludes FDG-PET. Therefore, external generalizability is demonstrated only partially and the multi-tracer generalization claim should be qualified.

Response:

We thank the reviewer for highlighting the need to clarify the scope of our external validation.

In this study, both AV45-PET and FDG-PET synthesis were evaluated internally on the ADNI dataset, whereas external validation on the independent OASIS-3 dataset was conducted only for

AV45-PET. As described in the original manuscript (Section 4.3), FDG-PET scans in OASIS-3 were not included in the external evaluation due to the lack of a standardized PUP processing pipeline.

We agree that the current OASIS-3 results provide evidence of cross-dataset generalizability for AV45-PET synthesis only and cannot be directly extended to FDG-PET synthesis. Accordingly, we will clarify the scope of external validation and refine the relevant statements regarding multi-tracer generalizability in the revised manuscript to ensure that our conclusions accurately reflect the evaluations performed.

Action in Manuscript:

We have revised the manuscript accordingly across the following sections:

- **Abstract: Results, Line 66**
- **Introduction: Summary of Contributions, Line 81**
- **Results: Cross-Dataset Generalizability on OASIS-3, Line 106**
- **Figure 2 Caption, Line 112—-114**

Revised Text in Manuscript (Abstract: Results, Line 66):

"Extensive evaluations on multi-center datasets demonstrate that DIReCT++ not only produces synthetic PET images (^{18}F-AV-45 and ^{18}F-FDG) of superior fidelity but also accurately recapitulates disease-specific patterns, with cross-dataset generalizability demonstrated for ^{18}F-AV-45 PET synthesis."

Revised Text in Manuscript (Introduction: Summary of Contributions, Line 81):

"Comprehensive experiments demonstrate that our model generates multi-tracer PET images with superior fidelity compared to state-of-the-art methods, with cross-dataset generalizability further demonstrated for ^{18}F-AV-45 PET synthesis."

Revised Text in Manuscript (Results: Cross-Dataset Generalizability on OASIS-3, Line 106):

"External validation on the OASIS cohort confirmed that the advantages of DIReCT++ in ^{18}F-AV-45 PET synthesis generalize beyond the training data (Fig. 2A)."

Revised Text in Manuscript (Results: Cross-Dataset Generalizability on OASIS-3, Line 106):

"This robust performance across diverse datasets affirms the state-of-the-art status of DIReCT++ especially for ^{18}F-AV-45 PET synthesis, highlighting its superior fidelity and generalization capacity."

Revised Text in Manuscript (Figure 2 Caption, Line 112—-114):

"(A) Radar plots show normalized quantitative results (SSIM, MSE, MAE, PSNR) for FDG-PET and AV45-PET reconstruction on ADNI, and cross-dataset generalization of AV45-PET on OASIS."

Revised Text in Manuscript (Figure 2 Caption, Line 112—-114):

"(C) Generalization to OASIS for AV45-PET using models trained on ADNI."

R2-8: Non-Significance vs. Formal Equivalence Testing

Comment R2-8:

> Non-significance does not prove equivalence: Using $p > 0.05$ between real and synthetic PET as evidence of high fidelity is statistically insufficient. Equivalence testing (e.g., TOST) with clinically meaningful equivalence margins is more appropriate.

Response:

We thank the reviewer for pointing out that the absence of a statistically significant difference does not constitute evidence of equivalence. We agree that relying solely on paired comparisons with $p > 0.05$ between real and synthetic PET images is statistically insufficient to support claims of high fidelity.

To provide a more comprehensive evaluation of regional uptake agreement, we performed additional statistical analyses using the originally preserved subject-level paired data. In addition to paired t-tests and multiple-comparison correction, we calculated the mean paired differences and their 95% confidence intervals, mean absolute error (MAE), Lin's concordance correlation coefficient (CCC), intraclass correlation coefficient (ICC), and Bland–Altman bias and limits of agreement. Together, these metrics provide a more comprehensive assessment of mean bias, regional errors, and subject-level agreement.

The additional analyses showed that, across the 98 brain regions evaluated for each tracer, the median regional CCC values were 0.9745 and 0.9450 for AV45 and FDG, respectively, while the corresponding median regional MAE values were 0.0160 and 0.0159. Under the predefined multiple-comparison correction procedure, no statistically significant mean paired differences were detected in the vast majority of regions and diagnostic groups.

Regarding the suggested two one-sided tests (TOST), we acknowledge the value of formal equivalence testing. However, the regional uptake values in our study were derived from per-image min–max intensity normalization rather than clinically standardized quantitative measurements such as SUV or SUVR. Currently, a well-established equivalence margin with a clear clinical interpretation is lacking for measurements on this normalized intensity scale. Defining an arbitrary equivalence margin without sufficient justification could lead to conclusions that lack clinical interpretability. Therefore, we did not perform TOST to claim clinical equivalence.

Accordingly, we will revise the statistical interpretation in the manuscript to clearly distinguish the absence of statistically significant mean differences, regional uptake agreement, and formal equivalence. We will also moderate the original statements that inferred high fidelity from $p > 0.05$. Furthermore, we will explicitly acknowledge in the Limitations section that our current analyses support agreement in normalized regional uptake values but do not establish clinical quantitative equivalence. Future work will explore PET quantitative evaluation scales with clear biological and clinical interpretations, the definition of appropriate equivalence margins, and corresponding formal equivalence testing methods.

Action in Manuscript:

We have revised the manuscript accordingly across the following sections:

- **Results Section 2.3: Regional Uptake Value Definition, Line 119**
- **Results Section 2.3: Objectives of Regional Statistical Analyses, Line 134**

- **Results Section 2.3: Real-Syn Regional Uptake Concordance, Line 136**
- **Methods Section 4.5.1: Real-Syn Pairwise Consistency Testing, Line 331**
- **Methods Section 4.5.1: AD/CN Between-Group Effect Sizes, Line 333**
- **Methods Section 4.5.1: False Discovery Rate (BH-FDR) Comparison Families, Line 336**
- **Discussion, Line 186**

Revised Text in Manuscript (Results Section 2.3: Regional Uptake Value Definition, Line 119):

"To assess the clinical validity of the synthesized PET images, we parcellated both synthetic and real PET scans into anatomical regions using `mri_synthseg` *(FreeSurfer v7.4.1) [1, 2], a deep learning-based segmentation tool that provides high-resolution cortical and subcortical labeling consistent with the Desikan–Killiany and aseg atlases. This approach yielded a total of 98 distinct regions of interest (ROIs), including cortical, subcortical, cerebellar, and brainstem structures. Regional normalized uptake values (arbitrary units) were computed as the mean voxel intensity within each ROI from PET images normalized using per-image min–max intensity scaling. Fig.* **??** *illustrates this analysis for key ROIs in two representative ADNI subjects, demonstrating strong regional consistency between synthetic and real images, as well as clear distinctions between syntheses from different diagnostic groups."*

Revised Text in Manuscript (Results Section 2.3: Objectives of Regional Statistical Analyses, Line 134):

"For a comprehensive quantitative evaluation, we performed two complementary regional statistical analyses on the ADNI cohort. First, we compared synthetic and real images to assess the normalized regional uptake agreement of DIReCT++. Second, we compared the AD and CN groups separately in real and synthetic PET to assess whether disease-related regional differences were retained."

Revised Text in Manuscript (Results Section 2.3: Real-Syn Regional Uptake Concordance, Line 136):

"Paired-sample t-tests between synthetic and real images revealed no statistically significant differences in normalized regional uptake across the principal brain regions shown in Fig. **??***A, B for the overall cohorts of both tracers ($p > 0.05$). Across all 98 ROIs, no overall comparisons remained significant after Benjamini–Hochberg false discovery rate (FDR) correction (all $q > 0.05$). Among the diagnosis-stratified paired comparisons, only the right pericalcarine cortex (ROI 2021) in the ^{18}F-FDG MCI subgroup showed a significant mean paired difference after FDR correction ($q = 0.0481$). The median concordance correlation coefficients (CCC) across the 98 ROIs were 0.9745 for ^{18}F-AV-45 and 0.9450 for ^{18}F-FDG, with corresponding median mean absolute errors (MAE) of 0.0160 and 0.0159 (normalized intensity units). For the 16 representative tracer–ROI combinations in Fig.* **??***A, B, descriptive regression slopes ranged from 0.9794 to 1.0127 (median, 1.0007) for ^{18}F-AV-45 and from 0.9844 to 1.0101 (median, 1.0001) for ^{18}F-FDG, without a consistent direction of deviation from unity. These results indicate close agreement in normalized regional uptake between synthetic and real PET, with detailed statistics for the representative ROIs provided in Supplementary Table* **??***."*

Revised Text in Manuscript (Methods Section 4.5.1: Real-Syn Pairwise Consistency

Testing, Line 331):

"

- ***Intra-group consistency**: We performed paired two-sided t-tests to compare normalized regional uptake between real and Syn-PET images within each ROI, both across the overall cohorts and within individual diagnostic groups. The null hypothesis was that the mean paired difference in normalized regional uptake was zero. Mean paired differences with 95% confidence intervals, mean absolute error (MAE), Lin's concordance correlation coefficient (CCC), the two-way absolute-agreement single-measure intraclass correlation coefficient (ICC(A,1)), and Bland–Altman bias and 95% limits of agreement were also calculated to assess regional agreement. For the 16 representative tracer–ROI combinations shown in Fig.* **??***A, B, we additionally performed subject-level ordinary least-squares regression of synthetic on real normalized regional uptake in the overall cohorts, reporting slopes, intercepts, and their unadjusted 95% confidence intervals.*

"

Revised Text in Manuscript (Methods Section 4.5.1: AD/CN Between-Group Effect Sizes, Line 333):

"

- ***Inter-group discriminability**: To assess whether Syn-PET preserves diagnostic signal, we conducted independent two-sided Student's t-tests (assuming equal variances) comparing mean normalized regional uptake between the AD and CN groups in the same ROIs—first using real PET and then repeating the analysis with Syn-PET. Cohen's d was calculated separately for real and synthetic PET as the AD–CN mean difference divided by the pooled within-group sample standard deviation, to characterize the direction and magnitude of the group differences.*

"

Revised Text in Manuscript (Methods Section 4.5.1: False Discovery Rate (BH-FDR) Comparison Families, Line 336):

"*All t-tests were two-tailed, and p-values were adjusted for multiple ROI comparisons using the Benjamini–Hochberg false discovery rate (FDR) procedure [3]. For the paired real–synthetic comparisons, FDR correction was applied across 98 ROIs separately for each tracer and analysis group (overall, AD, CN, EMCI, LMCI, and MCI), yielding 12 families (2 tracers × 6 analysis groups) and 1,176 tests in total. For the AD–CN comparisons, FDR correction was applied across 98 ROIs separately for each tracer and PET type (real or synthetic), yielding four families and 392 tests in total. Statistical significance was defined as an FDR-adjusted* $q < 0.05$*."*

Revised Text in Manuscript (Discussion, Line 186):

"*Several limitations of this study present clear avenues for future work. First, the current text guidance relies on fundamental patient information and imaging knowledge. Incorporating additional clinically accessible data, such as blood biomarkers [4], could further enhance subject-level precision, necessitating the development of targeted cross-modal pretraining or prompt-learning strategies. Second, while DIReCT++ synthesizes* ^{18}*F-AV-45 and* ^{18}*F-FDG, extending the framework to include tau-PET imaging would provide a more comprehensive biomarker profile for sensitive screening at the preclinical stage. Third, to maximize clinical utility, future iterations could integrate downstream task guidance (e.g., diagnosis or prognosis labels) directly into the generative model, enabling controllable*

synthesis [5, 6] tailored for specific clinical decisions. Regional uptake agreement was evaluated using per-image min–max-normalized intensities rather than standardized SUV or SUVR, and clinically interpretable equivalence margins have not yet been established for these measurements. Future work will explore clinically meaningful PET quantification and justified equivalence margins to enable formal equivalence testing. Finally, the evaluation was conducted on well-characterized but limited research cohorts (ADNI, OASIS). Prospective validation on more diverse, real-world clinical populations (including data from thick-slice or low-field MRI [7]) is essential to fully establish the generalizability and robustness of the framework."

R2-9: FDR Multiplicity Correction Families and Reporting

Comment R2-9:

> Multiple-comparison reporting should be clearer: FDR correction is appropriate, but the statistical family, number of tests, and adjusted p/q-values should be reported explicitly.

Response:

We thank the reviewer for the valuable suggestion regarding the transparency of multiple-comparison correction reporting. We agree that, although the original manuscript stated that false discovery rate (FDR) correction was applied, the statistical families, number of tests, and adjusted p/q-values were not sufficiently specified.

To address this concern, we have confirmed the original regional statistical data and performed a systematic re-evaluation of the regional comparisons between real and synthetic PET images.

For the paired regional uptake agreement analysis between real and synthetic PET, we performed two-sided paired t-tests across all 98 regions of interest (ROIs) for each tracer, both in the overall cohorts and within individual diagnostic subgroups. Benjamini–Hochberg FDR correction was applied separately within each statistical family, comprising 12 families and 1,176 regional comparisons in total. The corresponding results for representative brain regions in the overall cohorts, including raw p-values and FDR-adjusted q-values, have been compiled in a new supplementary table.

For the AD versus CN comparisons shown in Figure 4C–F, we additionally performed two-sided independent-samples t-tests separately for AV45 Real, AV45 Syn, FDG Real, and FDG Syn PET. Benjamini–Hochberg FDR correction was applied across all 98 ROIs within each of these four statistical families, comprising 392 comparisons in total.

The re-evaluation showed that all 20 significant annotations in Figure 4C/D remained statistically significant after FDR correction, while all 20 non-significant annotations in Figure 4E/F remained non-significant. Sensitivity analyses using Welch's t-test did not change the significance classifications of these regions. Thus, the existing significant and non-significant classifications in Figure 4C–F are supported by the FDR-adjusted results and do not require changes.

In the revised manuscript, we will explicitly report the statistical families and corresponding numbers of tests, clarify the FDR correction procedure and significance thresholds in the Statistical Analysis section and the relevant descriptions of Figure 4, and provide adjusted q-values for representative brain regions to improve the transparency and reproducibility of the statistical analysis.

Action in Manuscript:

We have revised the manuscript accordingly across the following sections:

- **Results Section 2.3: AD/CN Diagnostic Differentiation and Effect Sizes, Line 138**
- **Methods Section 4.5.1: AD/CN Between-Group Effect Sizes, Line 333**
- **Methods Section 4.5.1: False Discovery Rate (BH-FDR) Comparison Families, Line 336**
- **Figure 4 Caption, Line 143—-146**

Revised Text in Manuscript (Results Section 2.3: AD/CN Diagnostic Differentiation and Effect Sizes, Line 138):

"Critically, independent-sample t-tests demonstrated that the significant differences between diagnostic groups observed in real PET data were also observed in the synthetic images (Fig. ??C, D; figs. ?? and ??). For instance, DIReCT++ recapitulated the statistically significant differences between AD and CN groups in the left precuneus for ^{18}F-AV-45 (real: $q = 7.73 \times 10^{-66}$; synthetic: $q = 1.25 \times 10^{-63}$; Fig. ??C) and in the left hippocampus for ^{18}F-FDG (real: $q = 4.27 \times 10^{-15}$; synthetic: $q = 1.27 \times 10^{-13}$; Fig. ??D). Similarly, regional differences that were non-significant in real PET data remained non-significant in the synthetic images after FDR correction (all $q \geq 0.05$ for the regions shown in Fig. ??E, F). The AD–CN effect-size directions were consistent between real and synthetic PET for all ten regions shown in Fig. ??C, D, although the absolute Cohen's d decreased in nine regions (maximum reduction, 0.0771). These findings support the preservation of group-level disease-related regional uptake patterns, while indicating variations in effect-size magnitude. Complete effect-size estimates and FDR-adjusted results for the displayed regions are provided in Supplementary Table ??."

Revised Text in Manuscript (Methods Section 4.5.1: AD/CN Between-Group Effect Sizes, Line 333):

"

- ***Inter-group discriminability**: To assess whether Syn-PET preserves diagnostic signal, we conducted independent two-sided Student's t-tests (assuming equal variances) comparing mean normalized regional uptake between the AD and CN groups in the same ROIs—first using real PET and then repeating the analysis with Syn-PET. Cohen's d was calculated separately for real and synthetic PET as the AD–CN mean difference divided by the pooled within-group sample standard deviation, to characterize the direction and magnitude of the group differences.*

"

Revised Text in Manuscript (Methods Section 4.5.1: False Discovery Rate (BH-FDR) Comparison Families, Line 336):

"All t-tests were two-tailed, and p-values were adjusted for multiple ROI comparisons using the Benjamini–Hochberg false discovery rate (FDR) procedure [3]. For the paired real–synthetic comparisons, FDR correction was applied across 98 ROIs separately for each tracer and analysis group (overall, AD, CN, EMCI, LMCI, and MCI), yielding 12 families (2 tracers × 6 analysis groups) and 1,176 tests in total. For the AD–CN comparisons, FDR correction was applied across 98 ROIs separately for each tracer and PET type (real or synthetic), yielding four families and 392 tests in total. Statistical significance was defined as an FDR-adjusted $q < 0.05$."

Revised Text in Manuscript (Figure 4 Caption, Line 143—-146):

*"**Caption: Regional comparison of mean normalized uptake between real and MRI-synthetic PET images.** (A,B) Violin plots show regional mean normalized uptake (arbitrary units) across cortical and subcortical regions for real and synthetic PET, comparing AV45 (A) and FDG (B). Statistical significance was assessed for each region. (C–F) AD–CN regional comparisons. Violin plots display mean normalized uptake in AD (red) and CN (green) subjects for selected regions. Each region uses paired half-violins: left (solid) = real PET, right (dashed) = synthetic PET; colors match across modalities. Solid lines indicate real PET significance, dashed lines indicate synthetic PET significance. The statistical annotations in panels (C–F) are consistent with the revised BH-FDR analysis: **** corresponds to $q < 10^{-4}$ and ns to $q \geq 0.05$. (C,D) show representative regions with significant AD–CN differences for AV45 and FDG, whereas (E,F) show selected regions without significant AD–CN differences after FDR correction. Cblm-WM denotes Cerebellum-White-Matter. Additional regional agreement and effect-size analyses are provided in the Supplementary Materials.*
"

R2-10: Subject-Level Agreement, Calibration, and Regional Concordance

Comment R2-10:

> PET agreement metrics should be strengthened: SSIM, PSNR, MSE, and MAE are useful image-similarity metrics but are insufficient for biomarker agreement. ICC/CCC, Bland-Altman analysis, regional SUV bias, and calibration analyses would improve clinical validation.

Response:

We thank the reviewer for the valuable suggestion to strengthen the evaluation of PET biomarker agreement. We agree that image-similarity metrics such as SSIM, PSNR, MSE, and MAE alone are insufficient to characterize subject-level regional uptake agreement and systematic differences between synthetic and real PET images.

To address this concern, we performed additional analyses using the original subject-level paired data across all 98 regions of interest (ROIs) for each tracer. Specifically, we calculated Lin's concordance correlation coefficient (CCC), the intraclass correlation coefficient (ICC(A,1)), mean paired differences with 95% confidence intervals, and Bland–Altman bias and 95% limits of agreement. These complementary metrics provide a more comprehensive assessment of regional errors, subject-level agreement, and systematic differences. Across the 98 ROIs, the median regional CCC values were 0.9745 for AV45 and 0.9450 for FDG.

Furthermore, to examine the relationship between synthetic and real regional uptake values, we conducted subject-level ordinary least-squares regression analyses for the 16 representative tracer–ROI combinations shown in Figure 4A/B. Real PET regional uptake was used as the independent variable and synthetic PET regional uptake as the dependent variable. Regression slopes, intercepts, and their corresponding 95% confidence intervals were calculated.

The regression slopes ranged from 0.9794 to 1.0127 (median: 1.0007) for AV45 and from 0.9844 to 1.0101 (median: 1.0001) for FDG. These representative regions did not exhibit a consistent pattern of linear compression or expansion, although modest deviations from a unit slope were observed in some regions. The regression results provide additional information on how synthetic regional uptake varies with real uptake levels and complement the CCC, ICC, and Bland–Altman analyses.

Importantly, the regional uptake measurements in this study were derived from per-image min–

max normalized PET intensities rather than clinically standardized quantitative SUV or SUVR measurements. Therefore, the reported agreement and regression results characterize normalized regional uptake only and should not be interpreted as evidence of clinical SUV/SUVR agreement, absolute quantitative accuracy, or clinically validated calibration. We will explicitly clarify this distinction and its implications in the revised manuscript. Future work will investigate PET synthesis and validation approaches that preserve clinically meaningful quantitative uptake measurements.

In the revised manuscript, we will expand the regional agreement supplementary table to include regression slopes, intercepts, and their 95% confidence intervals for the 16 representative tracer–ROI combinations, alongside the existing agreement and bias metrics. We will also clarify the corresponding statistical methods and limitations.

Action in Manuscript:

We have revised the manuscript accordingly across the following sections:

- **Results Section 2.3: Regional Uptake Value Definition, Line 119**
- **Results Section 2.3: Real-Syn Regional Uptake Concordance, Line 136**
- **Methods Section 4.5.1: Real-Syn Pairwise Consistency Testing, Line 331**
- **Discussion, Line 186**

Revised Text in Manuscript (Results Section 2.3: Regional Uptake Value Definition, Line 119):

"To assess the clinical validity of the synthesized PET images, we parcellated both synthetic and real PET scans into anatomical regions using `mri_synthseg` *(FreeSurfer v7.4.1) [1, 2], a deep learning-based segmentation tool that provides high-resolution cortical and subcortical labeling consistent with the Desikan–Killiany and aseg atlases. This approach yielded a total of 98 distinct regions of interest (ROIs), including cortical, subcortical, cerebellar, and brainstem structures. Regional normalized uptake values (arbitrary units) were computed as the mean voxel intensity within each ROI from PET images normalized using per-image min–max intensity scaling. Fig.* **??** *illustrates this analysis for key ROIs in two representative ADNI subjects, demonstrating strong regional consistency between synthetic and real images, as well as clear distinctions between syntheses from different diagnostic groups."*

Revised Text in Manuscript (Results Section 2.3: Real-Syn Regional Uptake Concordance, Line 136):

"Paired-sample t-tests between synthetic and real images revealed no statistically significant differences in normalized regional uptake across the principal brain regions shown in Fig. **??***A, B for the overall cohorts of both tracers ($p > 0.05$). Across all 98 ROIs, no overall comparisons remained significant after Benjamini–Hochberg false discovery rate (FDR) correction (all $q > 0.05$). Among the diagnosis-stratified paired comparisons, only the right pericalcarine cortex (ROI 2021) in the ^{18}F-FDG MCI subgroup showed a significant mean paired difference after FDR correction ($q = 0.0481$). The median concordance correlation coefficients (CCC) across the 98 ROIs were 0.9745 for ^{18}F-AV-45 and 0.9450 for ^{18}F-FDG, with corresponding median mean absolute errors (MAE) of 0.0160 and 0.0159 (normalized intensity units). For the 16 representative tracer–ROI combinations in Fig.* **??***A, B, descriptive regression slopes ranged from 0.9794 to 1.0127 (median, 1.0007) for ^{18}F-AV-45 and from 0.9844 to 1.0101 (median, 1.0001) for ^{18}F-FDG, without a consistent direction of deviation from*

unity. These results indicate close agreement in normalized regional uptake between synthetic and real PET, with detailed statistics for the representative ROIs provided in Supplementary Table **??***."*

Revised Text in Manuscript (Methods Section 4.5.1: Real-Syn Pairwise Consistency Testing, Line 331):

"

- ***Intra-group consistency****: We performed paired two-sided t-tests to compare normalized regional uptake between real and Syn-PET images within each ROI, both across the overall cohorts and within individual diagnostic groups. The null hypothesis was that the mean paired difference in normalized regional uptake was zero. Mean paired differences with 95% confidence intervals, mean absolute error (MAE), Lin's concordance correlation coefficient (CCC), the two-way absolute-agreement single-measure intraclass correlation coefficient (ICC(A,1)), and Bland–Altman bias and 95% limits of agreement were also calculated to assess regional agreement. For the 16 representative tracer–ROI combinations shown in Fig.* **??***A, B, we additionally performed subject-level ordinary least-squares regression of synthetic on real normalized regional uptake in the overall cohorts, reporting slopes, intercepts, and their unadjusted 95% confidence intervals.*

"

Revised Text in Manuscript (Discussion, Line 186):

"Several limitations of this study present clear avenues for future work. First, the current text guidance relies on fundamental patient information and imaging knowledge. Incorporating additional clinically accessible data, such as blood biomarkers [4], could further enhance subject-level precision, necessitating the development of targeted cross-modal pretraining or prompt-learning strategies. Second, while DIReCT++ synthesizes ^{18}*F-AV-45 and* ^{18}*F-FDG, extending the framework to include tau-PET imaging would provide a more comprehensive biomarker profile for sensitive screening at the preclinical stage. Third, to maximize clinical utility, future iterations could integrate downstream task guidance (e.g., diagnosis or prognosis labels) directly into the generative model, enabling controllable synthesis [5, 6] tailored for specific clinical decisions. Regional uptake agreement was evaluated using per-image min–max-normalized intensities rather than standardized SUV or SUVR, and clinically interpretable equivalence margins have not yet been established for these measurements. Future work will explore clinically meaningful PET quantification and justified equivalence margins to enable formal equivalence testing. Finally, the evaluation was conducted on well-characterized but limited research cohorts (ADNI, OASIS). Prospective validation on more diverse, real-world clinical populations (including data from thick-slice or low-field MRI [7]) is essential to fully establish the generalizability and robustness of the framework."*

R2-11: Classification Uncertainty and Confidence Intervals

Comment R2-11:

11. Classification uncertainty is underreported: ACC, AUC, sensitivity, and specificity should be accompanied by fold-wise mean ± SD or 95% confidence intervals rather than isolated point estimates.

Response:

We thank Reviewer 2 for this insightful and constructive observation.

Action in Manuscript:

Revised Text in Manuscript (Section X.X, Page Y):
“”

R2-12: Statistical Significance Testing of Classification Baselines

Comment R2-12:

12. Claims of statistical superiority require direct tests: Statements such as “significantly outperforming” should be supported by formal paired comparisons, such as DeLong testing for AUC or paired bootstrap/McNemar tests where appropriate.

Response:

We thank Reviewer 2 for this insightful and constructive observation.

Action in Manuscript:

Revised Text in Manuscript (Section X.X, Page Y):
“”

R2-13: One-Step Sampling Terminology and Rectified Flow Integration

Comment R2-13:

> One-step sampling terminology requires correction: The phrase “exact one-step sampling” is too strong for a distilled end-point predictor. The manuscript should distinguish exact ODE integration from learned one-step approximation.

Response:

We thank the reviewer for pointing out the imprecision in our description of one-step sampling. We agree that the continuous ODE process underlying rectified flow should be distinguished from the learned one-step approximation employed in DIReCT++.

Specifically, DIReCT++ generates the target PET image directly from the input MRI in a single forward pass, which does not imply exact integration of the underlying ODE. Accordingly, we will revise the expression “exact one-step sampling” to more accurately describe the learned one-step generation process.

Action in Manuscript:

We have revised the manuscript accordingly across the following sections:

- **Methods Section 4.2.3: Theoretical Formulation of One-Step Sampling, Line 283**
- **Methods Section 4.2.3: Formulation of Eq. (7), Line 286**

Revised Text in Manuscript (Methods Section 4.2.3: Theoretical Formulation of One-Step Sampling, Line 283):

"A key advantage of rectified flow is its potential to support efficient one-step generation through distillation."

Revised Text in Manuscript (Methods Section 4.2.3: Formulation of Eq. (7), Line 286):

"Mathematically, this learned one-step approximation is expressed as:"

R2-14: VLM Guidance and Adapter Implementation Details

Comment R2-14:

14. Domain-adapted VLM details are insufficient: The manuscript should report embedding dimensionality, PET image-feature extraction, prompt construction, affine-adapter training, threshold τ selection, and optimization details to ensure reproducibility.

Response:

We thank Reviewer 2 for this insightful and constructive observation.

Action in Manuscript:

Revised Text in Manuscript (Section X.X, Page Y):

""

R2-15: Equation 1, Inter-Tracer Similarity vs. Distinctness, and Tau

Comment R2-15:

> Equation (1) needs stronger justification: The semantic-similarity constraint between FDG and AV45 embeddings may conflict with the stated goal of preserving inter-tracer differences. A sensitivity analysis for τ and an explicit ablation are recommended.

Response:

We thank the reviewer for the valuable comments regarding the rationale and parameter sensitivity of the cross-tracer semantic similarity constraint. We agree that its design motivation and its role in balancing text–PET feature alignment and tracer-specific representations warrant further clarification.

In DIReCT++, we employ a pre-trained BiomedCLIP model to encode the textual descriptions of FDG and AV45, followed by two independent affine adapters that align the corresponding textual embeddings with their respective PET image features. However, optimizing only the individual text–PET alignment objectives may cause the two adapted guidance vectors to become highly similar, potentially diminishing the inter-tracer distinctions present in the original textual representations. We therefore introduce a cross-tracer similarity constraint, not to maximize the similarity between the two representations, but to regulate their relative similarity during feature alignment.

Regarding the selection of τ=0.82, we would like to clarify its specific basis. Before affine adaptation, the cosine similarity between the original FDG and AV45 textual embeddings produced by the pre-trained BiomedCLIP model was 0.8205374, which rounds to 0.82. Accordingly, we selected τ=0.82 as the target cross-tracer similarity, using the original relationship between the two textual representations as a reference. This allows the adapters to align the textual embeddings with their corresponding PET image features while maintaining approximately the relative similarity originally encoded by BiomedCLIP. Thus, τ=0.82 was not an arbitrarily assigned value, but was determined from the representational characteristics of the pre-trained text encoder.

We also acknowledge that Equation (1) in the original manuscript presented a simplified inequality constraint to convey the underlying design principle of cross-tracer semantic consistency. In the actual adapter optimization, this principle was implemented using a differentiable squared penalty, '$100 \times (\cos(\text{z_FDG}, \text{z_AV45}) - \tau)^2$', which encourages the cosine similarity between the adapted tracer embeddings to approach the specified target τ. This formulation penalizes deviations both above and below the target, thereby preventing the two tracer representations from becoming increasingly similar without constraint during text–PET alignment. We will clarify the implemented optimization objective and the rationale for selecting τ in the revised manuscript.

To further validate this design, we conducted additional adapter-level parameter sensitivity and cross-tracer constraint ablation experiments. Using identical two-fold data splits, original text and PET features, initialization procedures, and optimization settings, we evaluated τ values of 0.70, 0.82, 0.90, and 0.98, together with an additional configuration in which the cross-tracer similarity penalty was completely removed.

The results demonstrate that τ effectively regulates the relative similarity between the adapted tracer guidance vectors. As τ increased from 0.70 to 0.98, the mean cosine similarity between the adapted FDG and AV45 embeddings increased from 0.7032 to 0.9807. At the originally adopted τ=0.82, the resulting similarities were 0.8213 and 0.8215 for the two folds, respectively, closely matching the specified target. In contrast, removing the cross-tracer penalty increased these similarities to 0.9900 and 0.9894, indicating that optimizing text–PET alignment alone caused the two guidance vectors to become highly similar.

Further analysis showed that the individual text–PET feature alignment also increased with τ. At τ=0.82, the mean alignment similarities were 0.9687 for FDG and 0.9661 for AV45. Although removing the cross-tracer constraint further increased these alignment values, it simultaneously brought the cosine similarity between the two adapted tracer embeddings close to 0.99. These findings indicate that maximizing individual text–PET alignment does not necessarily preserve the original representational distinctions between tracers. Instead, the cross-tracer similarity term provides a controllable mechanism for balancing individual feature alignment and inter-tracer representational separation.

Therefore, the rationale for selecting τ=0.82 is to use the original similarity between the BiomedCLIP textual representations as a reference, maintaining substantial text–PET feature alignment while preventing the two adapted guidance vectors from becoming excessively similar.

The additional experiments directly demonstrate the influence of τ on adapter-level representation learning and the relative geometry of the tracer embeddings. Since these analyses focus on the representation level rather than retraining PET synthesis models under different τ settings, we present them as supporting evidence for the parameter selection and similarity constraint mechanism, rather than as an optimization study of PET synthesis performance.

We will incorporate the rationale for τ=0.82, the actual optimization formulation, and the additional sensitivity and ablation results into the revised manuscript to improve the methodological clarity and reproducibility of our approach.

Action in Manuscript:

We have revised the manuscript accordingly across the following sections:

- **Methods: Domain-Adapted VLM Formulation, Line 229—-241**
- **Results: The DIReCT++ Framework Overview, Line 88**
- **Supplementary Materials: Adapter Ablation Study**

Revised Text in Manuscript (Methods: Domain-Adapted VLM Formulation, Line 229—-241):

"where α_1, β_1 are learnable scalar weights and α_2, β_2 are learnable bias vectors. We choose scalar scaling rather than full linear projection to minimize trainable parameters and avoid overfitting, given the limited size of medical imaging datasets. The optimal parameters are obtained by minimizing the following objective:

$$\begin{aligned} \mathcal{L}_align = \& - \frac{1}{N_f} \sum_i = 1^{N_f} \operatorname{sim}\left(Adapt_f(c_f), f_f, i\right) \\ \& - \frac{1}{N_a} \sum_j = 1^{N_a} \operatorname{sim}\left(Adapt_a(c_a), f_a, j\right) \\ \& + \lambda_cross \left[\operatorname{sim}\left(Adapt_f(c_f), Adapt_a(c_a)\right) - \tau\right]^2, \end{aligned} \tag{1}$$

where $\operatorname{sim}(\cdot,\cdot)$ denotes cosine similarity, $f_{f,i}$ and $f_{a,j}$ denote the normalized PET image features of individual FDG and AV45 training subjects, and N_f and N_a are the corresponding numbers of available features. We set $\lambda_{cross} = 100$ and $\tau = 0.82$, the latter corresponding to the cosine similarity of the original FDG and AV45 BiomedCLIP textual embeddings (0.8205) before affine adaptation. The cross-tracer penalty encourages the adapted guidance vectors to retain approximately this reference similarity while being aligned with their respective PET image features, thereby regulating their relative representational separation.

By solving (1), we obtain the modality-specific guidance vectors $Adapt_f(c_f)$ and $Adapt_a(c_a)$, which are subsequently used to steer the rectified flow translation process in a domain-informed manner."

Revised Text in Manuscript (Results: The DIReCT++ Framework Overview, Line 88):

"An additional adapter-level sensitivity analysis showed that varying τ from 0.70 to 0.98 changed the mean cross-tracer embedding similarity from 0.7032 to 0.9807. At the adopted $\tau = 0.82$, the similarity was 0.8214, compared with 0.9897 when the cross-tracer penalty was removed, demonstrating its role in regulating inter-tracer representational similarity (Table)."

Revised Text in Manuscript (Supplementary Materials: Adapter Ablation Study):

"

Caption: Adapter-level sensitivity analysis of the cross-tracer similarity constraint. *Values are descriptive means across two folds. FDG and AV45 alignment denote the mean cosine*

similarity between each adapted tracer embedding and the corresponding individual PET image features. The cross-tracer similarity is the cosine similarity between the two adapted tracer embeddings. The analysis used 1,856 QC-filtered ADNI subjects. Only training-fold PET features were used for adapter optimization, and held-out features were used solely for evaluation. No cross-term indicates complete removal of the cross-tracer penalty.

Setting	***Training alignment***		***Held-out alignment***		***Cross-tracer***
	FDG	*AV45*	*FDG*	*AV45*	*similarity*
$\tau = 0.70$	*0.9437*	*0.9410*	*0.9436*	*0.9410*	*0.7032*
$\tau = 0.82$	*0.9687*	*0.9661*	*0.9687*	*0.9661*	*0.8214*
$\tau = 0.90$	*0.9838*	*0.9813*	*0.9838*	*0.9812*	*0.9018*
$\tau = 0.98$	*0.9948*	*0.9923*	*0.9948*	*0.9923*	*0.9807*
No cross-term	*0.9952*	*0.9927*	*0.9952*	*0.9926*	*0.9897*

”

R2-16: Demographic Confounding and Conditioning Variables

Comment R2-16:

16. Demographic confounding needs control: ADNI groups differ significantly in age, sex, and other variables that are also used in the conditioning prompt. Matched, adjusted, or subgroup analyses are needed to quantify confounding effects.

Response:

We thank Reviewer 2 for this insightful and constructive observation.

Action in Manuscript:

Revised Text in Manuscript (Section X.X, Page Y):

“”

R2-17: Baseline Adaptation, Tuning, and Fair Comparison Protocols

Comment R2-17:

17. Baseline adaptation and fairness should be documented: Some comparator methods originate from segmentation or MRI-to-CT synthesis. The authors should clearly describe how each baseline was adapted and confirm identical data splits, preprocessing, tuning effort, and training budgets.

Response:

We thank Reviewer 2 for this insightful and constructive observation.

Action in Manuscript:

Revised Text in Manuscript (Section X.X, Page Y):

“”

R2-18: Computational Efficiency, Parameter Counts, and Hardware Benchmarks

Comment R2-18:

18. Computational-efficiency claims need direct evidence: The manuscript reports approximately 138 GPU-hours for training, but should also report inference time per 3D volume, parameter count, FLOPs, memory use, and direct runtime comparisons with diffusion/flow baselines.

Response:

We thank Reviewer 2 for this insightful and constructive observation.

Action in Manuscript:

Revised Text in Manuscript (Section X.X, Page Y):

“”

R2-19: Code, Seed, Split, and Model Weight Release

Comment R2-19:

19. Reproducibility is a strength but can be improved: Public code, trained models, software versions, and hardware details are valuable. The authors should additionally provide fixed data-split files, random seeds, and a repository commit/version used for the reported experiments.

Response:

We thank Reviewer 2 for this insightful and constructive observation.

Action in Manuscript:

Revised Text in Manuscript (Section X.X, Page Y):

“”

R2-20: Clinical Translation Scope and Translational Boundaries

Comment R2-20:

> Clinical translation claims should be moderated: Statements regarding widespread screening, clinically actionable biomarkers, and early intervention exceed the evidence from retrospective ADNI/OASIS cohorts. Prospective, diverse, multi-center clinical validation is needed before such claims are justified.

Response:

We appreciate the reviewer’s thoughtful suggestions regarding the clinical translation of our work.

One of the primary motivations of DIReCT++ is to explore the feasibility of synthesizing multi-tracer PET images from routine MRI, potentially alleviating the challenges associated with PET acquisition costs, radiation exposure, and limited accessibility. Our evaluations on the ADNI and OASIS-3 datasets provide preliminary evidence of the framework's PET synthesis performance, while regional uptake analyses and downstream classification on ADNI further support its potential clinical relevance.

Nevertheless, translating these findings from retrospective research cohorts into clinical practice requires more comprehensive evidence. We agree with the reviewer that the current results are insufficient to support the direct application of synthetic PET in routine clinical screening or treatment decision-making. Accordingly, we will moderate statements regarding widespread screening, clinically actionable biomarkers, and early intervention to better distinguish the demonstrated findings from potential future clinical applications. We will also clarify that prospective validation in more diverse, real-world clinical populations remains necessary.

Action in Manuscript:

We have revised the manuscript accordingly across the following sections:

- **Introduction, Line 81**
- **Discussion, Line 175**
- **Discussion, Line 177**
- **Discussion, Line 179**
- **Discussion, Line 188**

Revised Text in Manuscript (Introduction, Line 81):

"Our work proposes a translational approach to precision biomarker synthesis, offering a scalable and data-efficient strategy with potential applications in the early assessment of dementia."

Revised Text in Manuscript (Discussion, Line 175):

"Beyond its technical merits, the primary significance of DIReCT++ lies in its potential to improve access to biomarker-related imaging information, thereby addressing a major challenge in the early assessment of dementia."

Revised Text in Manuscript (Discussion, Line 175):

"By generating synthetic estimates of both amyloid and metabolic PET from a single routine MRI scan, our framework offers a scalable approach for multimodal imaging research."

Revised Text in Manuscript (Discussion, Line 175):

"The potential clinical relevance of this approach is supported by the observed consistency of disease-related regional uptake patterns between synthetic and real PET (Figs. 3 and 4), as well as by our results on personalized MCI diagnosis and stratification (Fig. 5B, C), where the combination of MRI and synthetic multi-tracer PET achieved performance comparable to MRI with real PET and outperformed models using MRI or real PET alone."

Revised Text in Manuscript (Discussion, Line 177):

"It could enable widespread screening roughly for individuals at the preclinical or prodromal stages in primary care settings, and provide a basis for exploring more accessible assessment strategies."

Revised Text in Manuscript (Discussion, Line 177):

"DIReCT++ thus offers a potential foundation for more accessible multi-modal imaging approaches in future clinical workflows."

Revised Text in Manuscript (Discussion, Line 179):

"DIReCT++ moves beyond this by achieving precise generation of subject-specific PET images that recapitulate disease-related regional uptake patterns."

Revised Text in Manuscript (Discussion, Line 188):

"By generating synthetic PET images that support diagnostic classification and discrimination between MCI stages, our work offers a potential approach to bridging the gap between the biological definition of AD and its clinical application."

Revised Text in Manuscript (Discussion, Line 188):

"DIReCT++ represents a step toward scalable, accessible, and precise tools for early assessment of neurodegenerative disorders."

Part 3: Response to Reviewer 3 — Major Comments

R3-1: Circularity in Prompt Conditioning and Diagnostic Label Encoding

Comment R3-1:

1. Circularity: the conditioning prompt contains the information that defines the diagnostic labels. This is my principal concern. The clinical prompt encodes "neuropsychological scores" (lines 314-315), and Table S1 lists MMSE, CDR, FAQ, GDS, and NPI-Q. In ADNI, diagnostic group membership is defined largely by MMSE, CDR, and Logical Memory II performance, and the authors state that EMCI/LMCI were assigned using Logical Memory II (lines 358-360). The synthetic PET is then fed to a classifier to predict those same labels. Any gain of "MRI + synthetic PET" over "MRI alone" could therefore reflect clinical scores passed through the generator, not molecular information inferred from anatomy.

The following are needed to resolve this:

The exact prompt template, and a list of which scores are included. Is Logical Memory II or CDR in it? The manuscript is inconsistent: line 95 says "age, gender, clinical scores," lines 314-315 say "age, sex, weight, and neuropsychological scores," and Fig. 1 shows only "Age, Gender, ...". A classifier using the clinical variables alone, and one using MRI + clinical variables, as reference baselines. If MRI + clinical variables matches MRI + synthetic PET, the synthetic PET adds nothing. Synthesis and classification without clinical text (tracer token only), and with shuffled or mismatched prompts. If synthetic PET changes substantially when a CN subject's MRI is paired with an AD patient's scores, the images reflect the prompt rather than the patient.

Response:

We express our deepest gratitude to Reviewer 3 for raising this foundational point.

Action in Manuscript:

Revised Text in Manuscript (Section X.X, Page Y):

""

R3-2: Table S1 Score Distributions, Plausibility, and Imputation Protocols

Comment R3-2:

2. Table S1 values appear implausible and may indicate imputation that encodes the label directly. Several within-group standard deviations are not credible for integer-valued clinical scales. Examples: EMCI MMSE 27.50 ± 0.50, LMCI MMSE 24.47 ± 0.50, LMCI NPI-Q 4.00 ± 0.03, EMCI GDS 1.01 ± 0.14, and LMCI GDS 3.02 ± 0.28. The LMCI mean MMSE (24.47) is also lower than both MCI-U (26.83) and typical ADNI LMCI values, and the CN weight SD (24.13 kg) is unusually large. The authors should explain how these values were computed and whether missing scores were imputed

with group means. If they were, the prompt literally encodes group membership, and the classification results in Fig. 5 would be invalid. Please report missingness per variable and per group, and the imputation strategy.

Response:

We express our deepest gratitude to Reviewer 3 for raising this foundational point.

Action in Manuscript:

Revised Text in Manuscript (Section X.X, Page Y):

“”

R3-3: Absence of Longitudinal Prognostic Endpoints

Comment R3-3:

“Prognostic stratification” is not supported by the analysis.

EMCI vs. LMCI is a cross-sectional severity distinction based on a memory cutoff, not a prognostic outcome. The manuscript also describes LMCI as indicating “proximity to dementia conversion” and MCI-U as lacking “clear progression status” (lines 360-362), which conflates the two concepts. To justify the title and the claims in lines 178-181 and 249-250, the authors should perform a genuine prognostic analysis: progressive vs. stable MCI (pMCI/sMCI) over a defined follow-up (e.g., 36 months), with time-to-conversion modeling (Cox regression, C-index). Otherwise, “prognostic” and “stratification” should be removed from the title and claims.

Response:

We thank the reviewer for further highlighting the important distinction between MCI stage discrimination and longitudinal prognostic prediction.

Our original motivation for relating EMCI/LMCI classification to disease progression was based on previous studies examining the clinical characteristics and subsequent conversion risks associated with these two MCI stages (Aisen et al., 2015; Lin et al., 2022; Petersen, 2021). Accordingly, we included EMCI-versus-LMCI classification as a downstream task to investigate the performance of synthetic PET in fine-grained MCI stage stratification.

We agree that EMCI and LMCI in our study were defined according to baseline memory impairment, and that we did not perform progressive-versus-stable MCI (pMCI/sMCI) classification or time-to-conversion analysis based on longitudinal follow-up outcomes. Therefore, the current experiments do not establish the model’s ability to predict longitudinal disease progression.

However, we would like to clarify that stratification in our study refers to grouping MCI patients according to their current degree of cognitive impairment, rather than stratification according to future disease-conversion risk. To avoid ambiguity, we will remove or revise unsupported prognostic claims throughout the manuscript and clarify that the task concerns MCI stage stratification. We will also revise the descriptions in the Methods section that characterize LMCI as indicating proximity to dementia conversion and MCI-U as lacking clear progression status, ensuring that the group definitions accurately reflect the cross-sectional classification criteria used in our study.

Accordingly, we will revise the manuscript title to "Precision Synthesis of Multi-Tracer PET via VLM-Modulated Rectified Flow for MCI Stage Stratification" to clarify the scope of the stratification task.

Action in Manuscript:

We have revised the manuscript accordingly across the following sections:

- **Methods Section 4.3: Diagnostic Cohort Specifications (EMCI, LMCI, MCI-U), Line 295**
- **Methods Section 4.3**
- **Methods Section 4.5.2: Stage Discrimination vs. Diagnostic Classification**
- **Methods Section 4.5.2: Stage Discrimination vs. Diagnostic Classification, Line 344**

Revised Text in Manuscript (Methods Section 4.3: Diagnostic Cohort Specifications (EMCI, LMCI, MCI-U), Line 295):

"To account for the heterogeneity of MCI, we further subclassified individuals based on delayed recall performance on the WMS-R Logical Memory II Story A task: Early MCI (EMCI; $n = 291$) denoted a milder impairment stage, whereas Late MCI (LMCI; $n = 152$) denoted a more severe impairment stage."

Revised Text in Manuscript (Methods Section 4.3):

"The remaining MCI participants not classified as EMCI or LMCI were grouped as MCI-unclassified (MCI-U; $n = 520$)."

Revised Text in Manuscript (Methods Section 4.5.2: Stage Discrimination vs. Diagnostic Classification):

"

0.0.1 Diagnosis and MCI Stage Discrimination

"

Revised Text in Manuscript (Methods Section 4.5.2: Stage Discrimination vs. Diagnostic Classification):

"To rigorously evaluate the clinical utility of synthesized PET (Syn-PET) images, we assessed their capacity to support diagnostic classification and MCI stage discrimination across the AD continuum."

Revised Text in Manuscript (Methods Section 4.5.2: Stage Discrimination vs. Diagnostic Classification):

"Specifically, we trained identical DenseNet classifiers˜
cite to discriminate between three clinically salient diagnostic pairs: (1) AD versus CN individuals, a canonical diagnostic boundary; (2) MCI versus CN, representing early disease detection; and (3) EMCI versus late MCI (LMCI), representing different degrees of memory impairment."

Revised Text in Manuscript (Methods Section 4.5.2: Stage Discrimination vs. Diagnostic Classification, Line 344):

"This comparative framework enables a direct assessment of whether Syn-PET preserves the discriminative biomarker information encoded in real PET, and whether multi-tracer Syn-PET enhances downstream diagnostic classification and MCI stage stratification beyond single-modality or MRI-only approaches."

R3-4: Non-Significant Tests vs. Subject-Level Concordance and Amyloid Status

Comment R3-4:

4. Non-significant paired t-tests do not demonstrate equivalence, and group-level agreement does not show subject-level precision. Lines 146-150 infer "high fidelity" from $p > 0.05$ in paired t-tests. Absence of a significant difference is not evidence of equivalence. The central claim of the paper is subject-specific precision (lines 56, 69-73), which group-mean comparisons cannot test: a model that outputs a diagnosis-conditional average image would pass every test in Fig. 4. The authors should provide:

per-subject agreement for regional SUVR (Pearson/Lin's concordance, ICC, Bland-Altman bias and limits of agreement), overall and within each diagnostic group; formal equivalence testing (e.g., TOST) with pre-specified margins; most importantly, amyloid-positivity agreement. Using a standard composite SUVR or Centiloid threshold, report sensitivity, specificity, and Cohen's κ for classifying real $A\beta+$ vs. $A\beta-$ from synthetic AV-45, stratified by CN, MCI, and AD.

The clinically decisive test is whether the model detects amyloid-positive cognitively normal individuals and amyloid-negative MCI. These are exactly the cases where MRI and clinical scores cannot reveal pathology, and where the authors argue the method is most useful (lines 60-61, 217-219).

Response:

We express our deepest gratitude to Reviewer 3 for raising this foundational point.

Action in Manuscript:

Revised Text in Manuscript (Section X.X, Page Y):

""

R3-5: Standardized SUVR Terminology, Min-Max Scaling, and Effect Sizes

Comment R3-5:

5. Quantitative definitions: "SUV" vs. SUVR and min-max normalization. All volumes are min-max normalized to [0, 1] per image (lines 388-389). Regional values are then called "SUV" and normalized to cerebellar gray matter or pons (lines 137-138), and plotted as "Mean Uptake" on a 0-1 axis (Fig. 4, S1-S6). Please clarify:

whether ratios were computed on the original PET or on min-max-normalized images (the latter loses quantitative meaning); the reference region for florbetapir (whole cerebellum is the ADNI standard); the correct terminology (SUVR, not SUV).

Please also verify that the FDR correction described in Section 4.5.1 was actually applied to the p-values shown in the figures, and report effect sizes (Cohen's d) for real vs. synthetic group differences. Preserved significance with a much smaller or larger effect size is not "precise preservation."

Response:
We express our deepest gratitude to Reviewer 3 for raising this foundational point.

Action in Manuscript:

Revised Text in Manuscript (Section X.X, Page Y):
""

R3-6: Subject-Level Splitting, Generator-Classifier Leakage, and Sample Sizes

Comment R3-6:
6. Data splitting and possible leakage between generator and classifier. The manuscript states 3-fold cross-validation for synthesis (line 114), two-fold cross-validation (line 391), and 3-fold for classification (line 161). Please clarify:

whether splits are at the subject level (ADNI subjects contribute multiple visits); whether synthetic PET used to test the classifier on a given subject was produced by a generator that did not see that subject in training; whether generator and classifier folds are nested or aligned; the maximum allowed interval between MRI and PET acquisitions; whether only florbetapir scans were used (ADNI-3/4 also use florbetaben); whether ADNI's pre-processed PET (co-registered, averaged, uniform resolution) was used, and how multi-scanner differences were handled; which subjects (including MCI-U) entered each classification task, with per-task sample sizes.

Response:
We express our deepest gratitude to Reviewer 3 for raising this foundational point.

Action in Manuscript:

Revised Text in Manuscript (Section X.X, Page Y):
""

R3-7: Classification Metrics, Confidence Intervals, and MRI-Only Baselines

Comment R3-7:
7. Classification reporting and claims of significance.

Methods lists ACC, SEN, and SPE (line 459), but Results and Fig. 5 report AUC. No confidence intervals or statistical tests are given, yet the text says "significantly outperforming" (lines 171, 215-216). Please report mean ± SD across folds, 95% CIs, and DeLong tests for AUC differences. EMCI vs. LMCI is imbalanced (291 vs. 152), and specificity is low (∼55–75% in Fig. 5C). Please

report balanced accuracy. The MRI-only baseline for MCI vs. CN appears near chance (ACC ≈ 51% in Fig. 5B), which is well below published MRI-only performance. This inflates the apparent benefit of synthetic PET. Please justify the classifier and training protocol, or use a stronger MRI baseline.

Response:
We express our deepest gratitude to Reviewer 3 for raising this foundational point.

Action in Manuscript:

Revised Text in Manuscript (Section X.X, Page Y):
“”

R3-8: Vision-Language Grounding, Numerical Encoding, and Adapter Logic

Comment R3-8:
8. The vision-language component is overstated relative to what is implemented.

The adapted tracer guidance is a fixed constant. $Adapt_f(c_f)$ and $Adapt_a(c_a)$ are pre-computed and fixed (lines 306-307), with a scalar scale and a bias vector. They are therefore equivalent to two learned tracer tokens. Describing this as “domain knowledge encoding” and “prompt learning” overstates its role. An ablation replacing it with a randomly initialized learned tracer embedding is needed. Numeric encoding by CLIP. Subject-specific information comes from a frozen BiomedCLIP text encoder applied to a prompt with numbers. CLIP-style text encoders are known to represent numeric values poorly. Please show that the embedding meaningfully separates, for example, MMSE 29 from 24 (embedding distance vs. score difference), and compare against a simple tabular encoder (an MLP on normalized variables). If the tabular encoder performs as well, the VLM framing is not justified. Image features and constraint logic. BiomedCLIP is a 2D model. Please explain how $\mathbb{E}[f_f]$ and $\mathbb{E}[f_a]$ were computed from 3D PET (which slices, which aggregation). The constraint $sim(Adapt_f, Adapt_a) \geq \tau$ enforces similarity between tracers, which contradicts the stated goal of “preserving inter-tracer differences” (lines 98-99, 277-278). The value of τ, the optimization method for the constrained problem, and a sensitivity analysis are all missing. Notation. Fig. 1B uses α and β, while Eq. 1 uses α_1, α_2, β_1, β_2.

Response:
We express our deepest gratitude to Reviewer 3 for raising this foundational point.

Action in Manuscript:

Revised Text in Manuscript (Section X.X, Page Y):
“”

R3-9: Systematic Multi-Component Ablation Study

Comment R3-9:

9. Ablation study is missing. The manuscript reports no ablation beyond the single-tracer DIReCT comparison. Please add, at minimum:

no text; tracer token only; clinical prompt only; frozen vs. adapted tracer embedding; VLM text encoder vs. tabular encoder; one-step distilled vs. multi-step ODE sampling; joint vs. separate tracer generation.

Section 4.4 describes 3D RF as an ablation "isolating the contribution of BiomedCLIP guidance," but it differs from DIReCT++ in several respects at once.

Response:
We express our deepest gratitude to Reviewer 3 for raising this foundational point.

Action in Manuscript:

Revised Text in Manuscript (Section X.X, Page Y):
""

R3-10: Fair Comparisons Against Identically-Conditioned Baselines

Comment R3-10:

10. Baselines and fairness of comparison.

CycleGAN is an unpaired method applied to a paired task, and Swin-UNet is a segmentation architecture. Neither is a strong MRI-to-PET baseline. Directly relevant MRI-to-PET methods cited in the Introduction (e.g., refs. 11-14) are not compared. Most importantly, the baselines receive no clinical information, while DIReCT++ does. A fair comparison requires at least one strong baseline given the same clinical variables (e.g., 3D RF or a regression U-Net with a tabular conditioning branch). Please also clarify whether SegGuidedDiff was run in 2D or 3D, and how baseline hyperparameters were "optimized to their reported best configurations."

Response:
We express our deepest gratitude to Reviewer 3 for raising this foundational point.

Action in Manuscript:

Revised Text in Manuscript (Section X.X, Page Y):
""

R3-11: Quantitative Metric Reporting, Dispersion, and Exemplar Cases

Comment R3-11:

11. Image-quality metrics and figures. Radar plots normalized to the best method (Fig. 2A) hide absolute values and variance. Please provide:

a table of SSIM, PSNR, MAE, and MSE (mean ± SD), computed within a brain mask (background inflates PSNR/SSIM); paired significance tests against each baseline; per-tracer OASIS results; voxel-wise absolute error maps for representative and worst-case subjects; subject-level scatter plots of real vs. synthetic composite SUVR, colored by diagnosis and amyloid status; ROC curves with confidence bands.

The examples in Fig. 2B-C and Fig. 3 appear hand-selected; the selection criteria should be stated, or cases should be chosen at the median and 10th percentile of performance.

Response:
We express our deepest gratitude to Reviewer 3 for raising this foundational point.

Action in Manuscript:

Revised Text in Manuscript (Section X.X, Page Y):
“”

R3-12: Deterministic U-Net Comparison, Conditional Mean, and Uncertainty

Comment R3-12:

12. Determinism, conditional-mean behavior, and uncertainty. Because x_0 is the MRI (not noise) and inference is a single deterministic distilled step (Eq. 7), the model effectively performs conditional regression. For an admittedly ill-posed problem (lines 58-61), this tends to produce a conditional mean, which would smooth out individual pathology. Please:

compare against a plain regression U-Net with identical conditioning; discuss whether the model can represent uncertainty; consider providing uncertainty estimates (e.g., ensembles or stochastic sampling). For any clinical use, a synthetic amyloid image without a confidence measure is risky.

Also, the claim that rectified flow supports “exact one-step sampling” (lines 338-339) holds only for sufficiently straightened flows. The distillation procedure (teacher steps, loss, schedule after epoch 140) needs a full description.

Response:
We express our deepest gratitude to Reviewer 3 for raising this foundational point.

Action in Manuscript:

Revised Text in Manuscript (Section X.X, Page Y):
“”

R3-13: External Validation Limitations and OASIS-3 Prompt Specifications

Comment R3-13:

13. External validation is limited. OASIS-3 covers AV-45 only, and CN and AD only (no MCI), with notable age imbalance (66.3 vs. 72.8 years). Please state which prompt fields were available for OASIS (weight, FAQ, GDS, NPI-Q may be missing) and how gaps were handled; the Supplement mentions "secondary screening" without detail. External validation of FDG synthesis and of the MCI tasks would substantially strengthen generalizability claims.

Response:

We express our deepest gratitude to Reviewer 3 for raising this foundational point.

Action in Manuscript:

Revised Text in Manuscript (Section X.X, Page Y):

""

R3-14: Clinical Scope, Screening vs. Stratification, and 2024 Criteria

Comment R3-14:

Clinical positioning and overstatement.

The Discussion makes strong claims: "democratize access," "primary care screening," "qualitative leap," "new paradigm," and identifying candidates "most likely to benefit from disease-modifying therapies" (lines 208-234). Synthetic amyloid images cannot substitute for molecular confirmation, particularly for anti-amyloid therapy eligibility, where a false positive has real consequences. The authors should:

temper these claims;

discuss the 2024 revised Alzheimer's Association criteria, which define AD biologically by core biomarkers;

compare against, or at least discuss, plasma biomarkers such as p-tau217, which are now a low-cost, radiation-free route to amyloid status. They are mentioned only as a future prompt input (lines 236-237).

Response:

We thank the reviewer for the valuable suggestions regarding the clinical positioning of our study and the interpretation of its potential applications.

We agree that the clinical translation claims in the original manuscript should be moderated. Our study evaluated PET image synthesis, regional uptake patterns, and downstream classification performance using retrospective research cohorts, but did not directly establish the effectiveness of synthetic PET for routine clinical screening or treatment decision-making. Accordingly, we will revise the overstated claims regarding widespread clinical screening, identification of candidates for disease-modifying therapies, and transformative clinical impact. We will also clarify that synthetic PET cannot currently substitute for established molecular biomarker confirmation.

We also appreciate the reviewer's suggestion to contextualize our work within the latest AD diagnostic framework. The 2024 revised Alzheimer's Association criteria further emphasize the

biological definition of AD based on core biomarkers and recognize the diagnostic role of blood-based biomarkers, including plasma p-tau217 assays that meet appropriate performance requirements. We have expanded the Discussion to address this updated diagnostic framework, highlight the importance of these biomarkers in assessing AD pathology, and further clarify the current research positioning of DIReCT++.

Minor Comments

Action in Manuscript:

We have carefully verified this point. No further manuscript text modifications were required beyond clarifying the discussion.

R3-15: Incremental Methodological Novelty Beyond DIReCT (IPMI 2025)

Comment R3-15:

15. Incremental novelty over DIReCT (ref. 15). The main differences from the authors' own IPMI 2025 work appear to be the second tracer, the adapted tracer token, and distillation. The manuscript should state the delta explicitly and show, through ablation, which addition drives the gains.

Response:

We express our deepest gratitude to Reviewer 3 for raising this foundational point.

Action in Manuscript:

Revised Text in Manuscript (Section X.X, Page Y):

“”

Part 4: Response to Reviewer 3 — Minor Comments

We thank Reviewer 3 for the meticulous proofreading and valuable minor corrections. We have addressed every item in full:

[R3-minor-01] Grammar correction: 'This challenge stems':

> Line 57: "This challenges stems" → "This challenge stems."

Response & Action: We thank the reviewer for pointing out this grammatical error. We have corrected "This challenges stems" to "This challenge stems" in the revised manuscript.

Revised Text in Manuscript (Introduction, Line 77):

"This challenge stems from the inherent nature of MRI-to-PET synthesis as an ill-posed inverse problem; neurodegeneration visible on MRI is typically a downstream effect of prior molecular pathologies captured by PET˜
cite."

[R3-minor-02] Spelling correction: 'neurodegeneration':

> Lines 58-59: "neurodegenration" → "neurodegeneration."

Response & Action: We thank the reviewer for identifying this typographical error. We have corrected "neurodegenration" to "neurodegeneration" in the revised manuscript.

Revised Text in Manuscript (Introduction, Line 77):

"This challenge stems from the inherent nature of MRI-to-PET synthesis as an ill-posed inverse problem; neurodegeneration visible on MRI is typically a downstream effect of prior molecular pathologies captured by PET˜
cite."

[R3-minor-03] Standardization of SUV/SUVR terminology and plural forms:

> Line 137: "standardized uptake value (SUV) were" → "values (SUVRs) were."

Response & Action: Thank you for pointing out this grammatical and terminology issue. We agree that the original wording was imprecise. Upon reviewing the regional uptake analysis, we confirmed that the reported measurements represent mean regional intensities extracted from min–max-normalized PET images, rather than standardized uptake values (SUVs) or reference-region–normalized uptake ratios (SUVRs).

Therefore, directly replacing "SUV" with "SUVR" would not accurately reflect the measurements performed in this study. We will correct the grammatical error and use the more appropriate term "normalized regional uptake" in the revised manuscript, while ensuring consistent terminology and units in the corresponding figures.

Revised Text in Manuscript (Results: Recapitulation of Disease-Specific Biomarker Patterns, Line 119):

"Regional normalized uptake values were computed by extracting the mean voxel intensity within each ROI from min–max-normalized PET images."

[R3-minor-04] Explicit definitions of velocity field v_θ, parameters θ, and condition c:

> Line 301: "a learnable velocity field Vθ denotes the network parameters" is garbled; θ and c should be defined separately.

Response & Action: We thank the reviewer for identifying the unclear notation. We will clarify that
(V_
theta
) denotes the learnable velocity field and
(
theta
) its network parameters. We will also use
(
mathbf{c}
) consistently for the conditioning context, retaining its component-wise definition in the subsequent formulation.

Revised Text in Manuscript (Methods: Rectified Flow Formulation, Line 245):

"Specifically, given two heterogeneous medical distributions—$
mathbf0
sim
mathcal}$ and $
mathbf1
sim
*mathcal}$ (multi-tracer)—we model the transformation between them via a learnable velocity field $V_*
theta$, where $
theta$ denotes the network parameters."

[R3-minor-05] Consistency of diagnostic cohort naming between Methods and Results:

Section 4.5.1 describes paired tests for "AD and CN," but Results report CN, MCI, and AD (lines 145-147). Please reconcile.

Response & Action: We thank the reviewer for pointing this out. We have carefully revised this item in the manuscript.

[R3-minor-06] Anatomical specification of the temporoparietal junction ROI:

> Line 427: "Temporoparietal junction" is not a SynthSeg/Desikan-Killiany label. Please specify the ROIs actually used.

Response & Action: We thank the reviewer for pointing out the inconsistency between the anatomical terminology and the atlas labels used in our analysis. In the original manuscript, we used the general anatomical term "temporoparietal junction," which does not correspond to a distinct ROI label in the adopted atlas and may have caused ambiguity regarding the regions actually analyzed.

To clarify this issue, we re-examined the original ROI statistical analysis and Figure 4 plotting code

and confirmed the specific regions used. For FDG-PET, the selected regions included the bilateral hippocampus (Left/Right-Hippocampus), isthmus cingulate cortex (ctx-lh/rh-isthmuscingulate), and supramarginal gyrus (ctx-lh/rh-supramarginal). For AV45-PET, the selected regions included the bilateral precuneus (ctx-lh/rh-precuneus) and the inferior, middle, superior, and transverse temporal gyri (ctx-lh/rh-inferiortemporal, middletemporal, superiortemporal, and transversetemporal).

Accordingly, we will replace the imprecise or overly broad anatomical descriptions with the corresponding atlas labels and clearly distinguish the full ROI-based statistical analysis from the representative disease-related regions presented in Figure 4. These revisions will ensure that the Methods section accurately reflects the regions used in the analysis and the corresponding figures.

[R3-minor-07] Harmonization of FreeSurfer / SynthSeg citations and versions:

> Line 133 vs. line 390: "mri_synthseg (FreeSurfer v7.4.1)" and "SynthSeg

35

" cite different references for the same tool. Please use one version and one citation.

Response & Action: We thank the reviewer for pointing out the inconsistent nomenclature and citations for the brain parcellation method across the manuscript. We acknowledge that the use of "mri_synthseg (FreeSurfer v7.4.1)" and "SynthSeg" in different sections, together with inconsistent references, may have caused confusion.

To address this issue, we will consistently refer to the ROI parcellation method as SynthSeg and standardize its citation throughout the manuscript. We will also clearly distinguish the FreeSurfer-based skull-stripping step from SynthSeg-based brain parcellation and clarify that the ROI masks derived from T1-weighted MRI were used for regional PET analysis. These revisions will ensure a consistent and accurate description of the preprocessing and ROI analysis procedures.

Revised Text in Manuscript (Results Section 2.3: Regional Uptake Analysis, Line 119):

"To assess the clinical validity of the synthesized PET images, we parcellated the corresponding T1-weighted MRI scans using SynthSeg' [8]', a deep learning-based segmentation tool that provides high-resolution cortical and subcortical labeling consistent with the Desikan–Killiany and 'aseg' atlases, and applied the resulting ROI masks to both synthetic and real PET images."

Revised Text in Manuscript (Methods: Materials and Data Preprocessing, Line 301):

"For quantitative regional analysis, T1-weighted MRI scans were parcellated into 98 anatomical regions using SynthSeg' [8]', and the resulting ROI masks were applied to the spatially aligned PET images."

[R3-minor-08] Removal of duplicate reference entries (Refs. 27 and 33):

> Refs. 27 and 33 are duplicates (Fischl, FreeSurfer).

Response & Action: We thank the reviewer for identifying the duplicate reference. We have confirmed that References 27 and 33 refer to the same FreeSurfer publication. We will merge the duplicate entries and standardize the corresponding citations in the revised manuscript to ensure accuracy and consistency.

Revised Text in Manuscript (Results Section 2.3: Regional Uptake Analysis, Line 119):

"To assess the clinical validity of the synthesized PET images, we parcellated the corresponding T1-weighted MRI scans using SynthSeg' [8]', a deep learning-based segmentation tool that provides

high-resolution cortical and subcortical labeling consistent with the Desikan–Killiany and 'aseg' atlases, and applied the resulting ROI masks to both synthetic and real PET images."

[R3-minor-09] Author initial correction for Billot et al. (Ref. 35):

> Ref. 35: "Billot C" should be "Billot B."

Response & Action: We thank the reviewer for identifying the incorrect author name in Reference 35. We have verified that the first author is Benjamin Billot and will correct "Billot C" to "Billot B" in the revised manuscript.

Revised Text in Manuscript (References):

"35Ḃillot B, Greve DN, Puonti O, et al. SynthSeg: Segmentation of brain MRI scans of any contrast and resolution without retraining. Med. Image Anal. 2023;86:102789."

[R3-minor-10] Correction of author listing and title for Ref. 28:

> Ref. 28: author list and title should be verified.

Response & Action: We thank the reviewer for identifying the inaccuracies in Reference 28. Upon verification, we found that its author list, title, and publication details were inconsistent with the official SynthSeg publication. We will remove the incorrect entry and consistently cite the correct SynthSeg paper by Billot et al., published in Medical Image Analysis in 2023, to ensure the accuracy and consistency of the references.

Revised Text in Manuscript (References):

"Billot B, Greve DN, Puonti O, et al. SynthSeg: Segmentation of brain MRI scans of any contrast and resolution without retraining. Medical Image Analysis. 2023;86:102789. `https://doi.org/10.1016/j.media.2023.102789`*."*

[R3-minor-11] Harmonization of author name formatting in Ref. 26:

> Ref. 26: author formatting is garbled ("Xingchao Liu Chengyue Gong QL").

Response & Action: We thank the reviewer for identifying the author formatting error in Reference 26. We have verified that the correct authors are Xingchao Liu, Chengyue Gong, and Qiang Liu. We will correct the author list and update the reference to the officially published ICLR 2023 version to ensure accuracy and consistency.

Revised Text in Manuscript (References):

"Liu X, Gong C, Liu Q. Flow Straight and Fast: Learning to Generate and Transfer Data with Rectified Flow. International Conference on Learning Representations (ICLR). 2023."

[R3-minor-12] Correction of SPM citation details in Ref. 34:

> Ref. 34 (a gray-matter asymmetry protocol) is not an appropriate citation for SPM spatial normalization.

Response & Action: We thank the reviewer for pointing out the inappropriate citation for SPM. The original reference primarily describes voxel-wise gray matter asymmetry analysis rather than the SPM software used for spatial normalization, and it's our mistake. We will replace it with an authoritative reference for SPM and its associated methods to ensure that the citation accurately

reflects the neuroimaging preprocessing tool used in our study.

Revised Text in Manuscript (Methods: Materials and Data Preprocessing, Line 301):
"'Both modalities were jointly affinely normalized to the MNI152 standard space using SPM [9], with identical transformation matrices applied to preserve cross-modality alignment [10].'"

Revised Text in Manuscript (References (references.bib)):
"@bookspm, title = Statistical Parametric Mapping: The Analysis of Functional Brain Images, editor = Friston, Karl J. and Ashburner, John and Kiebel, Stefan J. and Nichols, Thomas E. and Penny, William D., publisher = Academic Press, year = 2007, isbn = 9780123725608

@articlefriston1995spatial, title = Spatial Registration and Normalization of Images, author = Friston, Karl J. and Ashburner, John and Frith, Christopher D. and Poline, Jean-Baptiste and Heather, John D. and Frackowiak, Richard S. J., journal = Human Brain Mapping, volume = 3, number = 3, pages = 165–189, year = 1995, doi = 10.1002/hbm.460030303 "

[R3-minor-13] Correction of citation for the FDG reference region in Ref. 36:
> Ref. 36 (an ADNI overview) is not an appropriate source for FDG AD-vulnerable regions.

Response & Action: We thank the reviewer for pointing out the inappropriate citation for AD-vulnerable regions in FDG-PET. We acknowledge that the original reference primarily describes the Alzheimer's Disease Neuroimaging Initiative (ADNI) and does not directly support the regional metabolic alterations discussed in our study.

To address this issue, we will replace the original citation with more relevant FDG-PET studies, including Mosconi et al. on hippocampal hypometabolism, He et al. on AD-related regional metabolic abnormalities, and Strom et al. on metabolic alterations involving the isthmus cingulate region. These references provide a more appropriate scientific basis for the selected disease-related regions.

Revised Text in Manuscript (Methods: Statistical Analysis (Regional Group-Wise Comparison), Line 326):
"Among the 98 anatomical ROIs, we selected representative disease-related regions for the comparisons shown in Fig. ??C,D.

For ^{18}F-FDG PET, these included the bilateral hippocampus (`Left-Hippocampus` and `Right-Hippocampus`), isthmus cingulate cortex (`ctx-lh-isthmuscingulate` and `ctx-rh-isthmuscingulate`), and supramarginal gyrus (`ctx-lh-supramarginal` and `ctx-rh-supramarginal`) [11–13].

For ^{18}F-AV45 PET, these included the bilateral precuneus (`ctx-lh-precuneus` and `ctx-rh-precuneus`) and the bilateral inferior, middle, superior, and transverse temporal gyri (corresponding to the `ctx-lh-` and `ctx-rh-` labels with the suffixes `inferiortemporal`, `middletemporal`, `superiortemporal`, and `transversetemporal`) [14]."

[R3-minor-14] Verification and correction of DDIM source citation in Ref. 25:
> Ref. 25 (MRI-to-CT for spinal segmentation) should be checked as the source for the 3D DDIM baseline.

Response & Action: We thank the reviewer for pointing out the inappropriate reference for the 3D DDIM baseline. We re-examined the original experimental code and confirmed that our baseline

was not based on the implementation by Graf et al. Instead, it was implemented using the 3D DiffusionModelUNet provided by MONAI GenerativeModels, with DDIMScheduler employed for accelerated sampling. We will remove the inappropriate citation to Graf et al. and replace it with the original DDIM paper by Song et al. and an appropriate reference for MONAI GenerativeModels. We will also clarify the implementation framework and sampling procedure of the 3D conditional diffusion baseline to ensure that the methodological description and references accurately reflect the experimental implementation.

Revised Text in Manuscript (Results: Superior Fidelity and Generalizability of Synthesized PET, Line 101):

"We rigorously evaluated the multi-tracer PET synthesis performance of DIReCT++ against a comprehensive suite of representative methods, including generative adversarial networks (CycleGAN [15]), transformer-based architectures (Swin-UNet [16]), and diffusion/flow-based models (SegGuidedDif [17], 3D DDIM [18], 3D RF [19]), alongside our baseline model, DIReCT [20], which performs mono-tracer generation without prompt adaptation."

Revised Text in Manuscript (Methods: Baseline Implementations and Competing Methods, Line 316):

"

- ***3D DDIM**: A 3D conditional diffusion baseline implemented using the `DiffusionModelUNet` and `DiffusionInferer` components of MONAI GenerativeModels (v0.2.3) [21], with `DDIMScheduler` [18] employed for accelerated sampling. This baseline was included to assess DIReCT++'s computational efficiency and sampling quality relative to accelerated diffusion approaches.*

"

[R3-minor-15] Data Availability disclosure of subject/image IDs and FDG accession:

> Data Availability mentions only "structural MRI and AV-45 PET"; FDG-PET should be added. Please also release the list of ADNI/OASIS subject and image IDs used, to allow exact replication.

Response & Action: We thank the reviewer for identifying the omission in the Data Availability statement and for emphasizing the importance of data traceability and reproducibility.

We acknowledge that FDG-PET was inadvertently omitted from the original statement. We will correct this omission and clarify the imaging modalities used in each dataset, distinguishing the multi-tracer PET data from ADNI and the AV45-PET data used for external validation on OASIS-3.

Regarding subject and image identifiers, we agree that detailed data selection and splitting information is important for reproducibility. We will organize the data selection criteria, fixed data splits, random seeds, and relevant reproducibility materials for the revised experiments and make them available to the extent permitted by the applicable ADNI and OASIS-3 data use agreements. Any sharing of subject- or image-level identifiers will comply with the corresponding data access and redistribution requirements.

We will also review and correct the data access and licensing information in the Data Availability statement to ensure consistency with the requirements of the respective data providers.

Revised Text in Manuscript (Data Availability, Line 380—-381):

"The data used in this study were obtained from two research datasets: the Alzheimer's Disease

Neuroimaging Initiative (ADNI) and the Open Access Series of Imaging Studies (OASIS-3). ADNI data are available to approved researchers through the LONI Image and Data Archive (IDA) at `https://ida.loni.usc.edu`*, subject to the ADNI Data Use Agreement. OASIS-3 data are available through the official OASIS data access process at* `https://sites.wustl.edu/oasisbrains/request-access/`*, subject to the corresponding Data Use Terms. The ADNI data used in this study included structural T1-weighted MRI,* ^{18}*F-FDG PET, and* ^{18}*F-AV45 PET, along with associated demographic and cognitive assessments. The OASIS-3 external validation used structural MRI and* ^{18}*F-AV45 PET.*

To facilitate reproducibility, the data selection criteria, fixed data splits, random seeds, and relevant implementation details for the revised experiments will be documented and made available to the extent permitted by the applicable data use agreements. Any sharing of subject- and image-level identifiers will comply with the corresponding data access and redistribution requirements."

[R3-minor-16] Comprehensive disclosure of network hyperparameters and architecture dimensions:

> Missing hyperparameters: τ, λ_F, λ_A, batch size, U-Net depth and channel widths, cross-attention placement, and parameter count.

Response & Action: We thank the reviewer for highlighting the insufficient reporting of model architecture details and training hyperparameters. We agree that providing these specifications is essential for methodological transparency and reproducibility.

We have further examined the original experimental notebooks and the configuration used for the revised experiments. The verified loss weights are λ_F = 1.0 and λ_A = 1.0, with a batch size of 1. The generator U-Net consists of four resolution levels with channel widths of (32, 64, 64, 128) and one residual block per level. Cross-attention is configured only at the deepest resolution level, with a conditioning dimension of 512. By instantiating the corresponding U-Net architecture, we verified that it contains 9,925,921 parameters, all of which are trainable. This count refers specifically to the generator U-Net and excludes the VLM and separate adaptation modules.

Regarding the similarity threshold τ, we verified that the revised E1 Full configuration uses $\tau = 0.82$ as a target in a soft penalty rather than as a strictly enforced constraint. We will therefore clarify the distinction between this implementation and the hard-constraint formulation presented in the original manuscript, and revise the mathematical description to accurately reflect the implemented optimization procedure.

In the revised manuscript, we will provide the verified architectural specifications and hyperparameters, clarify the placement of cross-attention, and explicitly define the scope of the reported parameter count. We will also distinguish settings confirmed from the original experimental records from those verified in the reconstructed experiments, ensuring that the reported implementation details are accurate and reproducible.

Revised Text in Manuscript (Methods: Rectified Flow Formulation, Line 263):

"This two-token context representation is injected into the velocity U-Net through cross-attention at its deepest resolution level, which is 512, allowing the corresponding features to attend to both the target imaging modality and the subject-specific clinical information. Critically, no additional trainable parameters are introduced for clinical metadata encoding—the clinical text embeddings are obtained from the frozen BiomedCLIP model."

Revised Text in Manuscript (Methods: Multi-Task Learning and One-Step Generation, Line 282):

"where $\mathbf{x}_{MRI}$ denotes the input magnetic resonance image, $\mathbf{y}_f$ and $\mathbf{y}_a$ are the corresponding ground-truth FDG and AV45 PET images (when available), and $\delta_F, \delta_A \in \{0, 1\}$ indicate the presence (1) or absence (0) of each PET modality for a given subject. The loss weights were set to $\lambda_F = 1.0$ and $\lambda_A = 1.0$, assigning equal weights to the two tracer-specific objectives. This formulation ensures that supervision is applied solely to available modalities, enabling training in heterogeneous clinical cohorts with incomplete multi-tracer imaging."

Revised Text in Manuscript (Methods: Implementation Details and Hardware, Line 347):

"All models were implemented in PyTorch (v2.6.0) and trained on NVIDIA RTX A6000 GPUs (48 GB memory per card). The Adam optimizer was used with an initial learning rate of
(2.5
times10{-5}
). Training was conducted for 165 epochs, with knowledge distillation applied from epoch 140 onward. Automatic Mixed Precision (AMP) was employed to accelerate training and reduce memory consumption. Each model required approximately 138 GPU hours for full convergence. A complete list of software dependencies and their versions is provided in Table S

software

for full reproducibility. Specifically, the generator employed a four-level 3D U-Net with channel widths of $(32, 64, 64, 128)$ and one residual block per level. Cross-attention was enabled only at the deepest resolution level, with a conditioning dimension of 512. The batch size was set to 1. The instantiated generator U-Net contained 9,925,921 parameters, all of which were trainable."

Revised Text in Manuscript (Methods > Domain-Adapted VLM, Line 229—-241):

"The adaptation parameters were optimized using a modality-specific alignment objective together with a soft penalty encouraging cross-tracer semantic consistency. In the revised E1 Full configuration, the similarity target was set to $\tau = 0.82$. Unlike a hard constraint, this penalty encourages the adapted tracer representations to maintain a specified level of semantic similarity without guaranteeing that the target is satisfied at every optimization step."

[R3-minor-17] Legend clarification for Fig. 1 ('Just Add It' mechanism):

> Fig. 1 legend: "Just Add It" should be replaced with a clear description (e.g., "element-wise addition").

Response & Action: We thank the reviewer for pointing out the unclear terminology in the legend of Figure 1. We agree that the phrase "Just Add It" is informal and does not adequately describe the underlying mathematical operation.

After reviewing the correspondence between the schematic and the one-step generation formulation, we confirmed that this operation represents element-wise addition of the input MRI and the predicted velocity field to obtain the synthesized PET image.

Accordingly, we will replace "Just Add It" with "Element-wise Addition" in the revised Figure 1 to provide a more precise and technically appropriate description of the operation.

Revised Text in Manuscript (Figure 1 Caption and Schematic):

"We have updated Figure 1 (revised-1029-fig1.pdf) by replacing the informal label "Just Add It" with "Element-wise Addition" to accurately depict the underlying mathematical operation."

[R3-minor-18] Fig. 4 C–F: Explicit clarification of MCI cohort comparisons:

> Fig. 4 C-F show only AD vs. CN. Please add MCI comparisons, given the paper's focus on MCI.

Response & Action: We thank the reviewer for the suggestion to include MCI comparisons in Figure 4C–F. We appreciate the importance of evaluating synthetic PET images in individuals with mild cognitive impairment.

The primary purpose of Figure 4C–F is to investigate whether synthetic PET images preserve representative AD-related regional differences observed in real PET images. We therefore selected AD and CN as contrasting groups to provide a clearer comparison of regional uptake patterns across distinct disease states. Given the clinical and pathological heterogeneity of MCI, whose imaging characteristics may overlap with those of both CN and AD, these panels were not intended to provide an exhaustive comparison across all diagnostic stages.

Instead, our evaluation of MCI focuses on the potential utility of synthetic PET for distinguishing cognitive impairment stages. Specifically, Figure 5 includes dedicated downstream classification tasks for MCI versus CN and EMCI versus LMCI, assessing the contribution of synthetic PET to MCI identification and the differentiation of early- and late-stage MCI. These analyses complement the AD-related regional comparisons presented in Figure 4C–F.

We acknowledge that the original manuscript did not sufficiently clarify this rationale. We will revise the relevant descriptions to explain the purpose of the AD/CN comparisons and the complementary role of the MCI-related classification experiments. We will also clarify that the overall regional analyses in Figure 4A–B included CN, MCI, and AD subjects without separate diagnostic-group stratification.

Revised Text in Manuscript (Results Section 2.3: Regional Statistical Analysis, Line 134):

"For a comprehensive quantitative evaluation, we performed two complementary regional analyses on the ADNI cohort. First, we compared synthetic and real PET images to assess their regional correspondence. Second, we compared the AD and CN groups using both real and synthetic PET images to examine whether representative AD-related regional differences observed in real PET were reflected in the synthesized images."

Revised Text in Manuscript (Results Section 2.3: Regional Statistical Analysis, Line 138):

"Critically, independent-sample t-tests demonstrated that the significant differences between diagnostic groups observed in real PET data were precisely preserved in the synthetic images (Fig. ??C, D; figs. ?? and ??). For instance, DIReCT++ recapitulated the statistically significant differences ($p < 0.001$) between AD and CN groups in the precuneus for ^{18}F-AV-45 (Fig. ??C) and in the hippocampus for ^{18}F-FDG (Fig. ??D). Similarly, regional differences that were non-significant in real PET data were also consistently non-significant in the synthetic images ($p > 0.05$, Fig. ??E, F). These findings confirm that DIReCT++ reliably preserves the essential biomarker information necessary for clinical differentiation. The AD–CN comparison was intentionally selected to illustrate representative AD-related regional differences between two distinct diagnostic states, rather than to

*provide an exhaustive comparison across all diagnostic groups. Given the heterogeneity of MCI, its diagnostic differentiation was evaluated separately through the downstream MCI-versus-CN and EMCI-versus-LMCI classification tasks (Fig. **??**B, C).”*

[R3-minor-19] Standardization of axis units and reference regions in figures:

Axis labels “Mean AV45/FDG Uptake” should read “SUVR,” with the reference region stated.

Response & Action: We thank the reviewer for pointing this out. We have carefully revised this item in the manuscript.

[R3-minor-20] Clean formatting of raw LaTeX symbols in submission system abstract:

> Abstract: The abstract in the submission system contains raw LaTeX commands (‘‘, ‘‘, ‘^{18}F‘), which should be cleaned.

Response & Action: We thank the reviewer for pointing out the incorrectly displayed LaTeX commands in the abstract within the submission system. We will correct this error when submitting the revised manuscript.

[R3-minor-21] Harmonization of sex vs. gender terminology across the text:

> Terminology: Please use “sex” consistently rather than alternating with “gender.”

Response & Action: We thank the reviewer for pointing out the inconsistent terminology. We will replace “gender” with “sex” in the relevant passage to ensure consistent terminology throughout the manuscript.

Revised Text in Manuscript (Results: Overview of the DIReCT++ Framework, Line 88):

“To achieve subject-wise precision, text guidance incorporating both common knowledge (e.g., general descriptions of PET tracers) and personal patient information (e.g., age, sex, clinical scores) is encoded by a pre-trained VLM and integrated into the RF model via cross-attention.”

Revised Text in Manuscript (Supplementary Materials):

“The demographic characteristics of this selected cohort, including age and sex distribution, are detailed in Table S2.”

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